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Oracle SQLcl: The Free Tool Every PL/SQL and APEX Developer Needs

APEX World - 26 & 27 March, 2026, Ede, NL
Thursday, 26 Mar 2026 11:00-11:50 am



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Oracle SQLcl (SQL Developer Command Line)

- SQLcl is SQL Developer just with CLI vs a GUI
- Several dozens of useful commands
- First, early adopter version was mentioned around 2014/2015
- "SQL*PLus++" , "SDSQL" *
- Modern replacement of SQL Plus
- Latest version 25.4.2 (February 2026)

<https://www.oracle.com/pl/database/sqldeveloper/technologies/sqlcl/>
<https://krisrice.io/2014-12-11-what-is-sdsq/>

```
ACCEPT
ALIAS
APEX
APPEND
AQ
ARBORI
ARCHIVE_LOG
ARGUMENT
AWR
BACKGROUND
BLOCKCHAIN_TABLE
BREAK
BRIDGE
BTITLE
CD
CERTIFICATE
CHANGE
CLEAR
CLOUDSTORAGE
CODESCAN
COLUMN
COMPUTE
CONNECT
CONNMGR
COPY
CS
CTAS
DATAPUMP
:|
```

About me

- Kielce, Poland - 200km (125 miles) south from Warsaw
- Oracle APEX and PL/SQL Developer
- Worldwide speaker since 2021
- Oracle ACE Pro

Find me at:

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[rgrzegorczyk](#) 

<https://rafal.hashnode.dev/>



Rafał Grzegorczyk



Providing what you **need**, not what you **want**

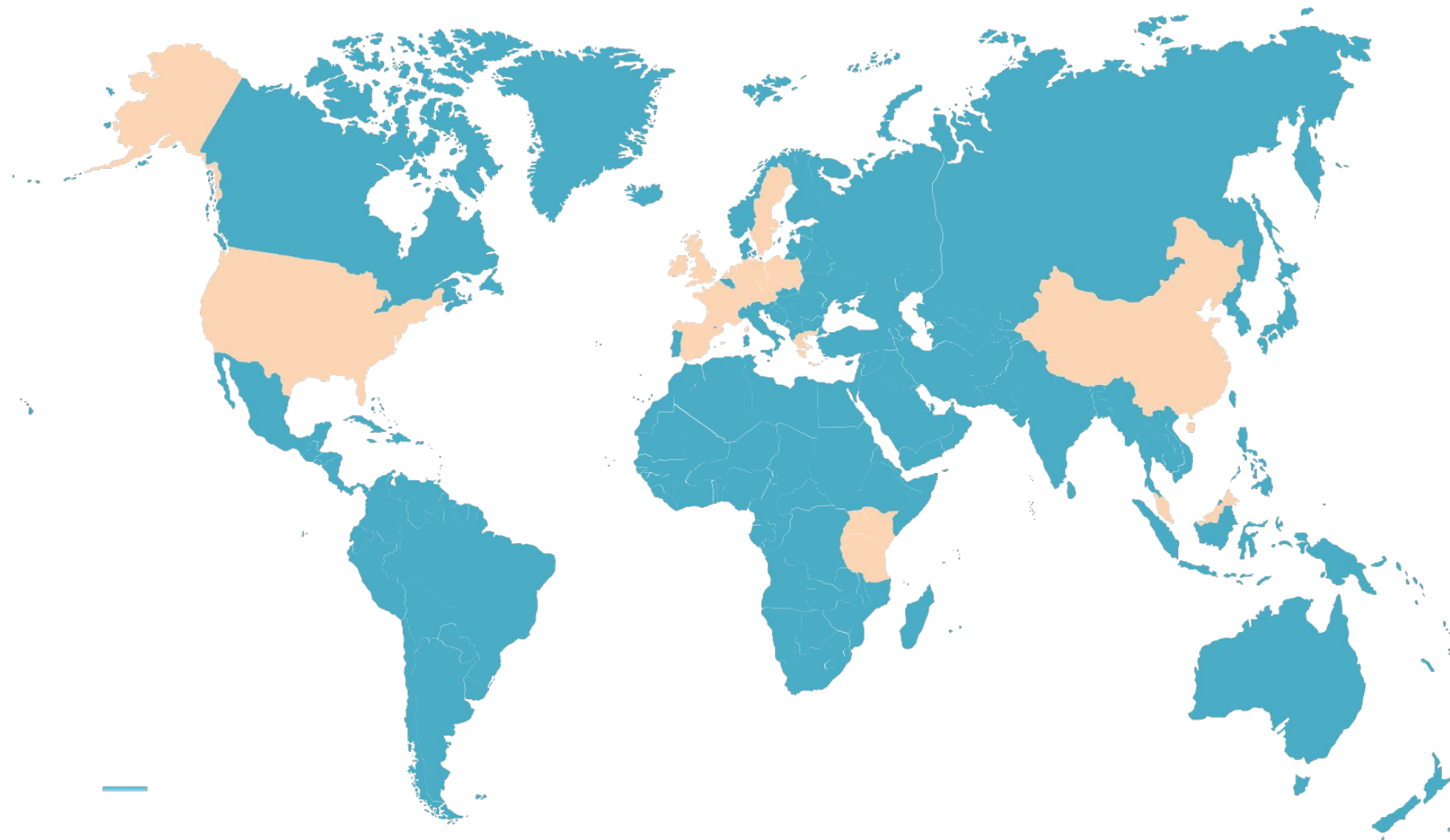


Scan to meet us
Pretius.com

300+
People

60+
APEX Devs

6
Oracle ACEs



70+
Customers

4
Continents

2
Warsaw, PL
London, UK



Partner





**GitHub - rgrzegorzcyk/conferences: A place
for public content related to my
presentations at various conferences**

github.com

<https://github.com/rgrzegorzcyk/conferences>



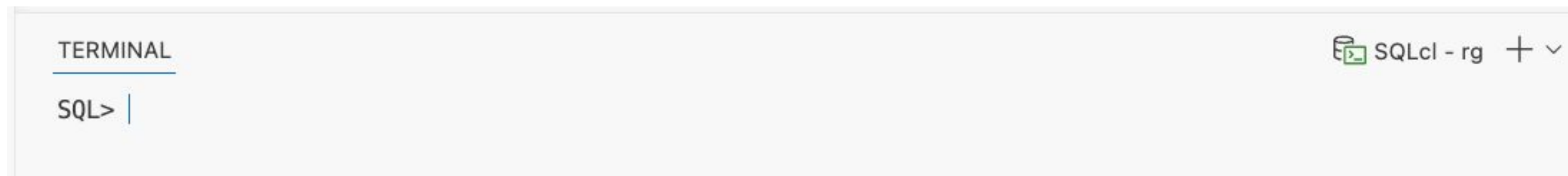
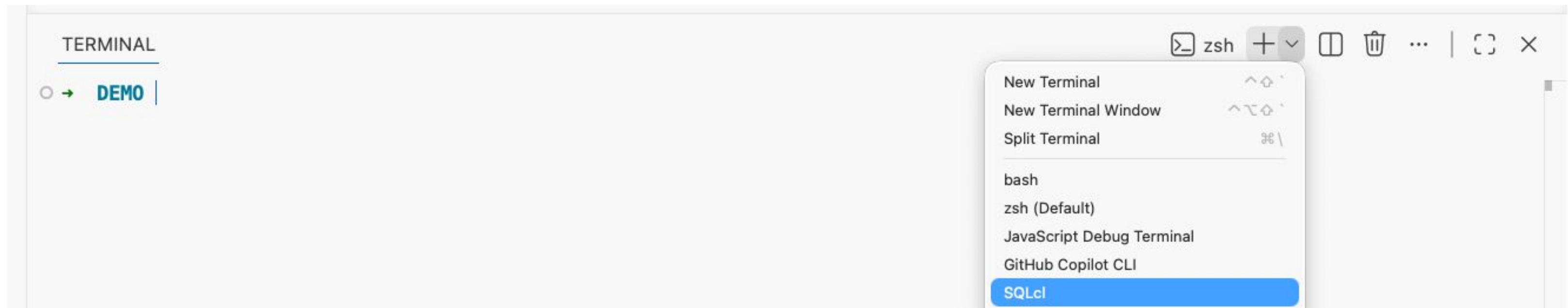
SQLcl features I find most useful:

1. Model Context Protocol (MCP) and AI support
2. Import/export data from/to files
3. Data Pump - export/import schemas
4. APEX export
5. CODESCAN - PL/SQL quality and performance checks
6. Projects - Database Application CI/CD
7. Liquibase

SQLcl installation



Do you have SQL Developer Extension in Visual Studio Code?
SQLcl is already there



For standalone version go
to **www.oracle.com/sqlcl**

SQLcl Downloads

Download the latest version of SQLcl now

Download

Get Started

Release Notes

Documentation

Forums

Download

Product Versions

25.4.2.044.1837

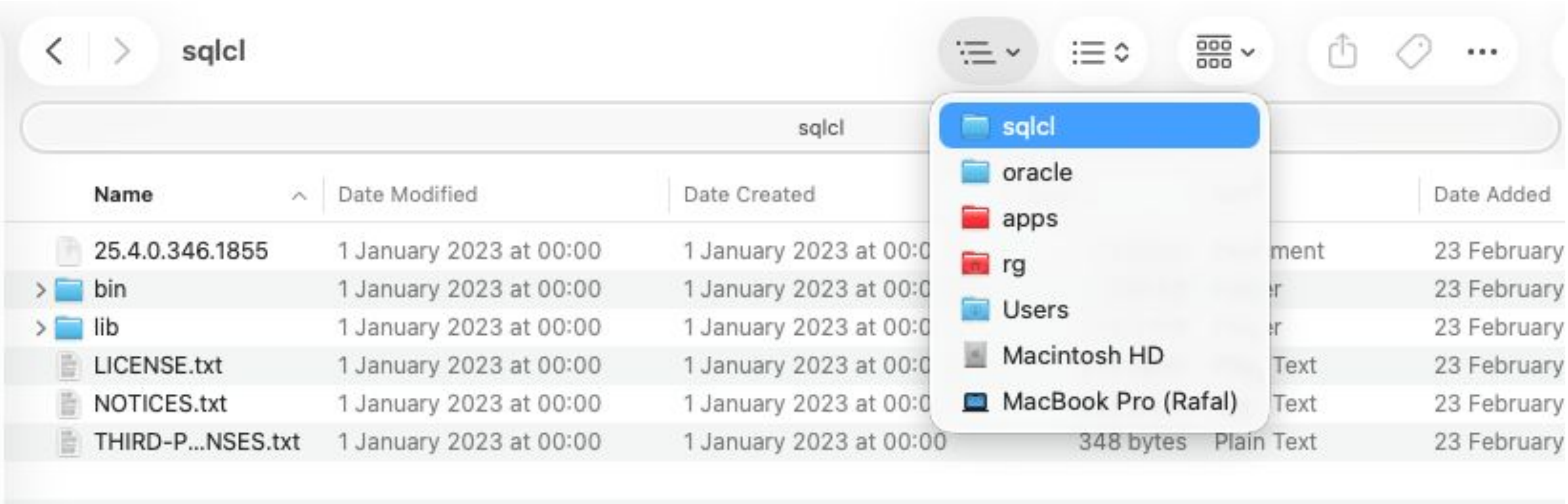
Related Resources

MD5: fb4536a793df5e03dc52d9078221f798

SHA1: dd0d80afe1ac8494565924b4e29c1949d1c31792

Oracle Free Use License

Unzip downloaded file and add to chosen folder



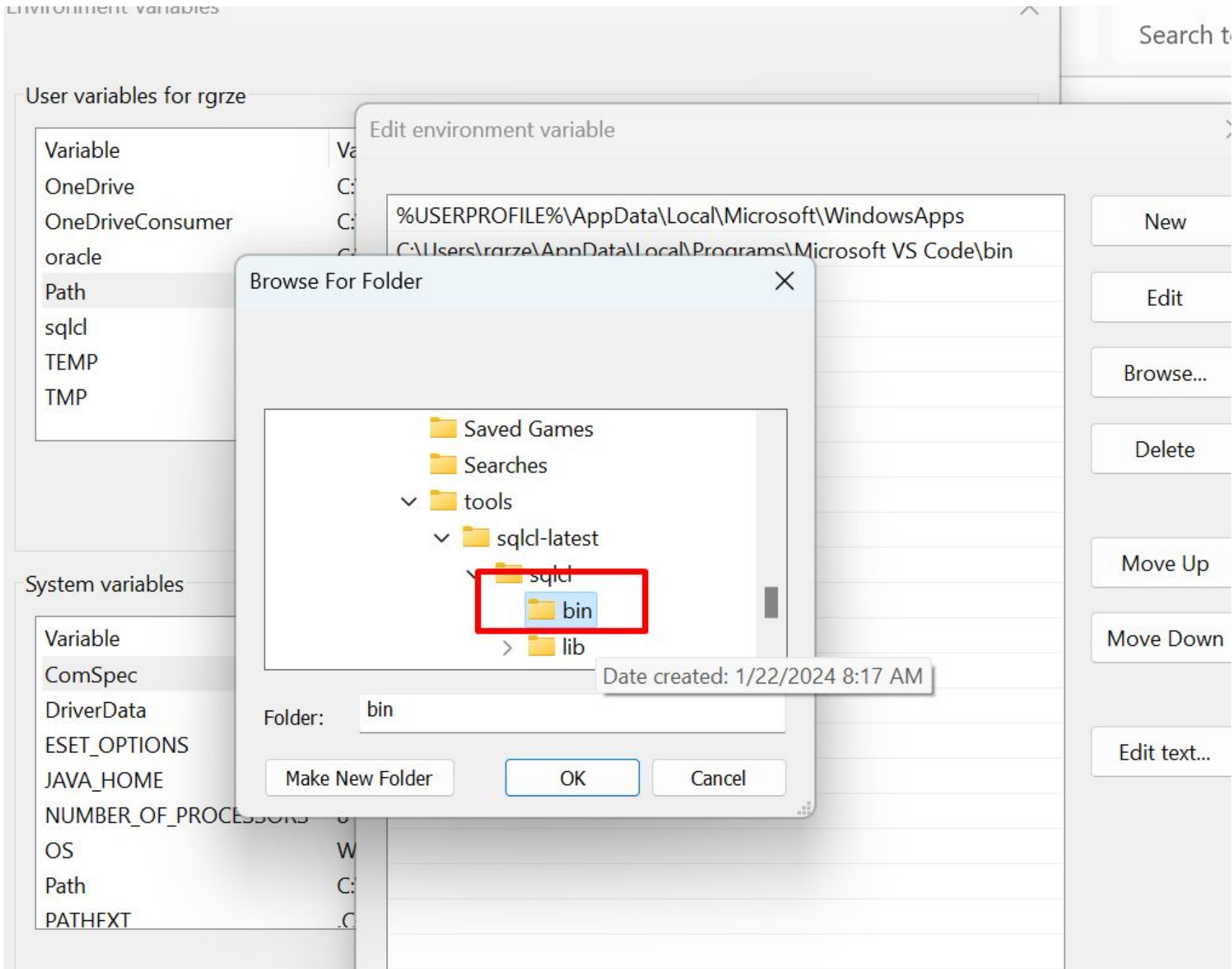
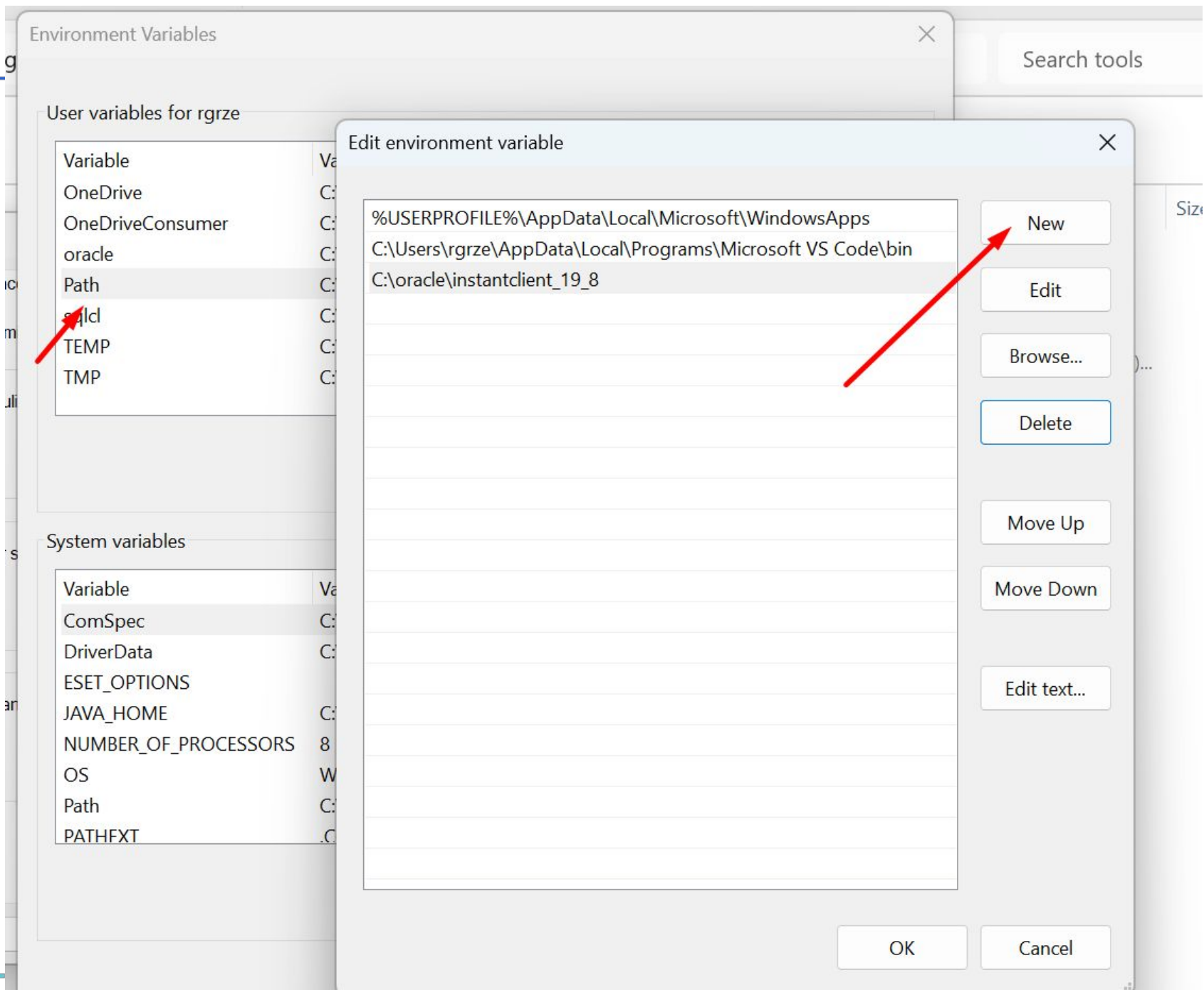


Add SQLcl "bin" folder to system path (**MacOS**)

Users/<your_user>/.zshrc (or .bash_profile)

```
export PATH="/Users/rg/apps/oracle/sqlcl/bin:$PATH"
```


Add SQLcl “bin” folder to system path (Windows)





Verify installation (JAVA is required)

> java -version

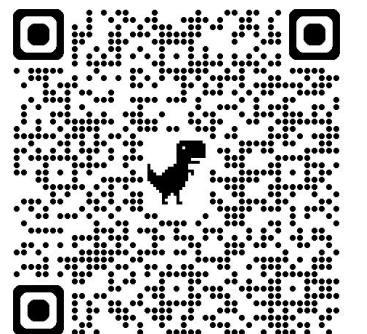
TERMINAL

```
● → DEMO java -version
java version "25.0.2" 2026-01-20 LTS
Java(TM) SE Runtime Environment (build 25.0.2+10-LTS-69)
Java HotSpot(TM) 64-Bit Server VM (build 25.0.2+10-LTS-69, mixed mode, sharing)
○ → DEMO
```

> sql -version

TERMINAL

```
● → DEMO sql -version
SQLcl: Release 25.4.0.0 Production Build: 25.4.0.346.1855
○ → DEMO
```



More about installation [here](#) ->

SQLcl

connect to database



Visual Studio Code SQL Developer Extension **(easiest way)**

EXTENSIONS: INSTALLED

@installed SQL Developer

Oracle SQL Developer ...

End-to-end development tools fo...

Oracle Corporation

117ms

Extension: Oracle SQL Developer Extension for VSCode

Oracle SQL Developer Extension for VSCode

Oracle Corporation | 758,601 | (115)

End-to-end development tools for your SQL and PL/SQL applications.

Disable

Uninstall

☒ Auto Update

DETAILS

FEATURES

CHANGELOG

Oracle SQL Developer Extension for VS Code

About the Oracle SQL Developer extension for VS Code

The gold standard for Oracle Database development and administration is now seamlessly integrated into Visual Studio Code

Installation

Identifier	oracle.sql-developer
Version	25.4.1
Last Updated	2 weeks ago
Size	493.92MB



Visual Studio Code SQL Developer Extension **(easiest way)**

SQL DEVELOPER

SQL SNIPPETS

CONNECTIONS

customers

priv

demo

Connection: dev_demo

Update Connection dev_demo

Connection Name

dev_demo

User Info

Proxy User

Authentication Type

Default

Role

Default

Username

DEMO

Password

.....

☒ Save Password

Connection Type

Cloud Wallet

Details

Proxy

Advanced

Configuration File

Choose File

Service

DEV_MEDIUM

Cancel

Test

Delete

Connect

Save

Details

Proxy

Advanced

Configuration File

Choose File

Service

DEV_MEDIUM

Cancel

Test

Test passed for connection: dev_demo

Source: Oracle SQL Developer Extension for VSCode

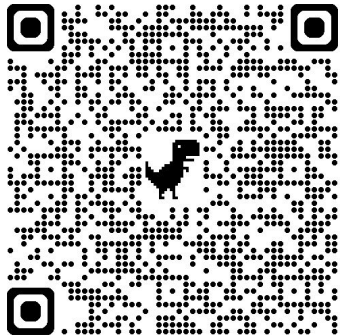
Cancel



Using SQLcl commands **(also easy)**

> conn -save

```
○ → DEMO git:(dev) x
```



More about connections [here](#) ->



> sql -name <connection_name>

TERMINAL

○ → DEMO sql -name dev_demo

SQLcl: Release 25.4 Production on Tue Mar 10 10:41:56 2026

Copyright (c) 1982, 2026, Oracle. All rights reserved.

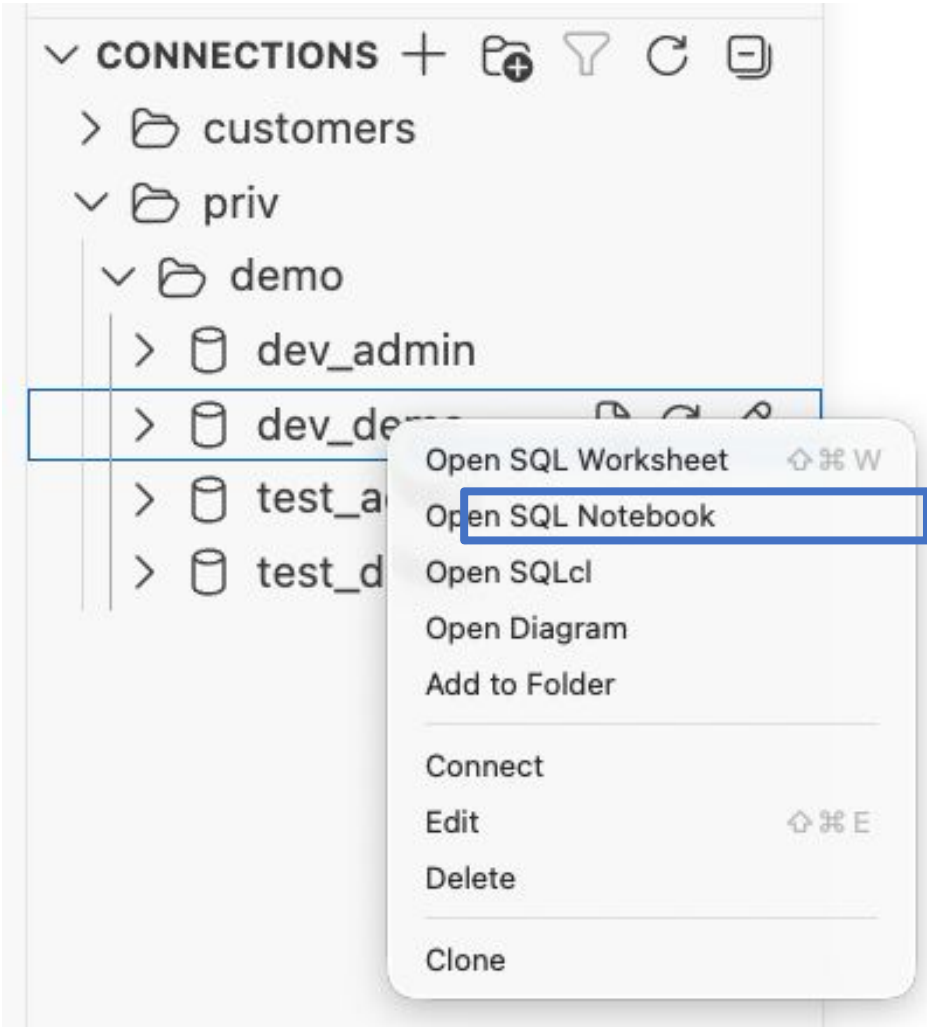
Connected to:

Oracle AI Database 26ai Enterprise Edition Release 23.26.1.2.0 – Production
Version 23.26.1.2.0

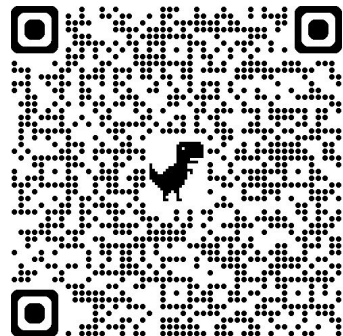
SQL>



Using VS Code SQL Developer extension



More about connections [here](#) ->



SQLcl

Export data



```
> set sqlformat insert
```

```
SQL> set sqlformat insert
SQL> select name, country, rating from eba_demo_ig_people fetch first 10 rows only;
REM INSERTING into EBA_DEMO_IG_PEOPLE
SET DEFINE OFF;
Insert into EBA_DEMO_IG_PEOPLE (NAME,COUNTRY,RATING) values ('Keizo Obuchi','Japan',22.475646);
Insert into EBA_DEMO_IG_PEOPLE (NAME,COUNTRY,RATING) values ('Antonín Novotný','Czech Republic',22.474758);
Insert into EBA_DEMO_IG_PEOPLE (NAME,COUNTRY,RATING) values ('Shivaji','India',22.474212);
Insert into EBA_DEMO_IG_PEOPLE (NAME,COUNTRY,RATING) values ('Frank Rijkaard','Netherlands',22.473873);
Insert into EBA_DEMO_IG_PEOPLE (NAME,COUNTRY,RATING) values ('Richard Pryor','United States',22.473754);
Insert into EBA_DEMO_IG_PEOPLE (NAME,COUNTRY,RATING) values ('Nico','Germany',22.473584);
Insert into EBA_DEMO_IG_PEOPLE (NAME,COUNTRY,RATING) values ('Alexander II of Scotland','United Kingdom',22.473236);
Insert into EBA_DEMO_IG_PEOPLE (NAME,COUNTRY,RATING) values ('Jacques Anquetil','France',22.472601);
Insert into EBA_DEMO_IG_PEOPLE (NAME,COUNTRY,RATING) values ('Hideki Shirakawa','Japan',22.47257);
Insert into EBA_DEMO_IG_PEOPLE (NAME,COUNTRY,RATING) values ('Chet Atkins','United States',22.472027);

10 rows selected.
```



```
> set sqlformat json
```

```
SQL> set sqlformat json
SQL> select name, country, rating from eba_demo_ig_people fetch first 10 rows only;
{"results":[{"columns":[{"name":"NAME","type":"VARCHAR2"}, {"name":"COUNTRY","type":"VARCHAR2"}, {"name":"RATING","type":"NUMBER"}], "items":
[
{"name":"Giuseppe Farina","country":"Italy","rating":22.153414}
,{"name":"Geena Davis","country":"United States","rating":22.152783}
,{"name":"Bengt I. Samuelsson","country":"Sweden","rating":22.152342}
,{"name":"Susan Hayward","country":"United States","rating":22.151325}
,{"name":"Grand Duchess Maria Nikolaevna of Russia","country":"Russia","rating":22.151002}
,{"name":"Chang Myon","country":"South Korea","rating":22.149127}
,{"name":"Vladimir Voronin","country":"Moldova","rating":22.148992}
,{"name":"P. W. Botha","country":"South Africa","rating":22.148361}
,{"name":"Hans Sloane","country":"United Kingdom","rating":22.146437}
,{"name":"Howard Aiken","country":"United States","rating":22.14543}
]
]
}}
10 rows selected.
```

```
> select /*insert*/ ( ... )
```

```
> select /*csv*/ ( ... )
```

```
> select /*json*/ ( ... ) also works
```




```
> set sqlformat csv
> spool [file_name]
> spool off
```

```
SQL> set feedback off
SQL> set sqlformat csv
SQL> spool my_csv_file.csv
SQL> select name, country, rating from eba_demo_ig_people fetch first 10 rows only;
"NAME","COUNTRY","RATING"
"Keizo Obuchi","Japan",22.475646
"Antonín Novotný","Czech Republic",22.474758
"Shivaji","India",22.474212
"Frank Rijkaard","Netherlands",22.473873
"Richard Pryor","United States",22.473754
"Nico","Germany",22.473584
"Alexander II of Scotland","United Kingdom",22.473236
"Jacques Anquetil","France",22.472601
"Hideki Shirakawa","Japan",22.47257
"Chet Atkins","United States",22.472027
SQL> spool off;
```

my_csv_file.csv X

my_csv_file.csv > data

1	"NAME","COUNTRY","RATING"
2	"Johann Sebastian Bach","Germany",30.167197
3	"Ludwig van Beethoven","Germany",30.109331
4	"Cleopatra VII of Egypt","Egypt",30.036677
5	"Galileo Galilei","Italy",29.96003
6	"Euclid","Unknown",29.884558
7	"Nero","Italy",29.859917
8	"Karl Marx","Germany",29.842897
9	"Marco Polo","Italy",29.771886
10	"Cicero","Italy",29.742594
11	"Vincent van Gogh","Netherlands",29.738438
12	



```
> set sqlformat json
> spool [file_name]
> spool off
```

```
SQL> set sqlformat json
SQL> spool my_json.json
SQL> select name, country, rating from eba_demo_ig_people fetch first 10 rows only;
{"results": [{"columns": [{"name": "NAME", "type": "VARCHAR2"}, {"name": "COUNTRY", "type": "VARCHAR2"}, {"name": "RATING", "type": "NUMBER"}], "items": [
{"name": "Johann Sebastian Bach", "country": "Germany", "rating": 30.167197}
, {"name": "Ludwig van Beethoven", "country": "Germany", "rating": 30.109331}
, {"name": "Cleopatra VII of Egypt", "country": "Egypt", "rating": 30.036677}
, {"name": "Galileo Galilei", "country": "Italy", "rating": 29.96003}
, {"name": "Euclid", "country": "Unknown", "rating": 29.884558}
, {"name": "Nero", "country": "Italy", "rating": 29.859917}
, {"name": "Karl Marx", "country": "Germany", "rating": 29.842897}
, {"name": "Marco Polo", "country": "Italy", "rating": 29.771886}
, {"name": "Cicero", "country": "Italy", "rating": 29.742594}
, {"name": "Vincent van Gogh", "country": "Netherlands", "rating": 29.738438}
]]}]SQL> spool off;
```

```
{ } my_json.json X
```

```
{ } my_json.json > ...
```

```
1  { "results": [ { "columns": [ { "name": "NAME", "type": "VARCHAR2" }, { "name": "CO
2  {
3  { "name": "Giuseppe Farina", "country": "Italy", "rating": 22.153414 }
4  , { "name": "Geena Davis", "country": "United States", "rating": 22.152783 }
5  , { "name": "Bengt I. Samuelsson", "country": "Sweden", "rating": 22.152342 }
6  , { "name": "Susan Hayward", "country": "United States", "rating": 22.151325 }
7  , { "name": "Grand Duchess Maria Nikolaevna of Russia", "country": "Russia
8  , { "name": "Chang Myon", "country": "South Korea", "rating": 22.149127 }
9  , { "name": "Vladimir Voronin", "country": "Moldova", "rating": 22.148992 }
10 , { "name": "P. W. Botha", "country": "South Africa", "rating": 22.148361 }
11 , { "name": "Hans Sloane", "country": "United Kingdom", "rating": 22.146437 }
12 , { "name": "Howard Aiken", "country": "United States", "rating": 22.14543 }
13 ] ] ]
14
```




- > set loadformat csv
- > unload sample_restaurant

```
SQL> set loadformat csv
SQL> unload sample_restaurant

format csv

column_names on
delimiter ,
enclosures ""
encoding UTF8
row_terminator default

** UNLOAD Start ** at 2026.03.17-08.46.50
Export Separate Files to /Users/rg/DEMO
DATA TABLE SAMPLE_RESTAURANT
File Name: /Users/rg/DEMO/SAMPLE_RESTAURANT_DATA_TABLE.csv
Number of Rows Exported: 11
** UNLOAD End ** at 2026.03.17-08.46.51
SQL> |
```

SAMPLE_RESTAURANT_DATA_TABLE.csv	
SAMPLE_RESTAURANT_DATA_TABLE.csv > data	
1	"ID","NAME","ADDRESS","RATING","LOGO_URL","HERO_URL","CATEGORY_ID","CREATED"
2	1,"ByteStart Breakfast Nook","Address #123","4.0","restaurant/bytestart-br
3	2,"SQLicious Bistro","Address #724","3.2","restaurant/sqlicious-bistro.jpg"
4	3,"PL/SQLovely Diner","Address #335","5.0","restaurant/sqlovely-diner.jpg"
5	4,"Oracool Cafe","Address #294","4.5","restaurant/oracool-cafe.jpg","resta
6	5,"APEXquisite Delights","Address #513","4.3","restaurant/apexquisite-deli
7	6,"SQLtastic Grill","Address #367","3.7","restaurant/sqltastic-grill.jpg","
8	7,"PL/SQL Paradise","Address #211","4.9","restaurant/sql-paradise.jpg","re
9	8,"APEXtraordinary Fare","Address #841","2.9","restaurant/apextraordinary-
10	9,"Oraculicious Cuisine","Address #751","4.7","restaurant/oraculicious-cui
11	10,"APEX Mex Grill","Address #571","5.0","restaurant/apex-mex-grill.jpg","
12	11,"SQL Sushi Bar","Address #382","4.7","restaurant/sql-sushi-grill.jpg","
13	



- > set loadformat json
- > unload sample_restaurant

```
SQL> set loadformat json
SQL> unload sample_restaurant

format json

** UNLOAD Start ** at 2026.03.17-08.51.04
Export Separate Files to /Users/rg/DEMO
DATA TABLE SAMPLE_RESTAURANT
File Name: /Users/rg/DEMO/SAMPLE_RESTAURANT_DATA_TABLE.json
Number of Rows Exported: 11
** UNLOAD End ** at 2026.03.17-08.51.04
SQL> 
```

```
{ } SAMPLE_RESTAURANT_DATA_TABLE.json X
{ } SAMPLE_RESTAURANT_DATA_TABLE.json > ...

1  {"results":[{"columns":[{"name":"ID","type":"NUMBER"}, {"name":"NAME"
2  [
3  {"id":1,"name":"ByteStart Breakfast Nook","address":"Address #123",'
4  ,{"id":2,"name":"SQLicious Bistro","address":"Address #724","rating'
5  ,{"id":3,"name":"PL\SQLovely Diner","address":"Address #335","rati
6  ,{"id":4,"name":"Oracool Cafe","address":"Address #294","rating":"4.
7  ,{"id":5,"name":"APEXquisite Delights","address":"Address #513","rat
8  ,{"id":6,"name":"SQLtastic Grill","address":"Address #367","rating":
9  ,{"id":7,"name":"PL\SQL Paradise","address":"Address #211","rating'
10 ,{"id":8,"name":"APEXtraordinary Fare","address":"Address #841","rat
11 ,{"id":9,"name":"Oraculicious Cuisine","address":"Address #751","rat
12 ,{"id":10,"name":"APEX Mex Grill","address":"Address #571","rating":
13 ,{"id":11,"name":"SQL Sushi Bar","address":"Address #382","rating":'
14 ]}]}
```

SQLcl

Import data



- > **load** [table_name] [file_name] [options]
- > **load** conference sample_restaurant_data_table.csv **NEW**

SAMPLE_RESTAURANT_DATA_TABLE.csv	
SAMPLE_RESTAURANT_DATA_TABLE.csv > data	
1	"ID", "NAME", "ADDRESS", "RATING", "LOGO_URL", "HERO_URL", "CATEGORY_ID", "CR
2	1, "ByteStart Breakfast Nook", "Address #123", "4.0", "restaurant/bytest
3	2, "SQLicious Bistro", "Address #724", "3.2", "restaurant/sqlicious-bist
4	3, "PL/SQLoely Diner", "Address #335", "5.0", "restaurant/sqloely-dine
5	4, "Oracool Cafe", "Address #294", "4.5", "restaurant/oracool-cafe.jpg",
6	5, "APEXquisite Delights", "Address #513", "4.3", "restaurant/apexquisit
7	6, "SQLtastic Grill", "Address #367", "3.7", "restaurant/sqltastic-grill
8	7, "PL/SQL Paradise", "Address #211", "4.9", "restaurant/sql-paradise.jp
9	8, "APEXtraordinary Fare", "Address #841", "2.9", "restaurant/apextraord
10	9, "Oraculicious Cuisine", "Address #751", "4.7", "restaurant/oraculicio
11	10, "APEX Mex Grill", "Address #571", "5.0", "restaurant/apex-mex-grill.
12	11, "SQL Sushi Bar", "Address #382", "4.7", "restaurant/sql-sushi-grill.



```
SQL> load conference sample_restaurant_data_table.csv NEW
```

```
csv
column_names on
delimiter ,
enclosures ""
double off
encoding UTF8
row_limit off
row_terminator default
skip_rows 0
skip_after_names
```

```
Create new table and load data into table DEMO.CONFERENCE
```

```
#INFO DATE format detected: DD-MON-RR
```

```
CREATE TABLE DEMO.CONFERENCE
(
  ID NUMBER(4),
  NAME VARCHAR2(128),
  ADDRESS VARCHAR2(26),
  RATING NUMBER(4, 1),
  LOGO_URL VARCHAR2(128),
  HERO_URL VARCHAR2(128),
  CATEGORY_ID NUMBER(3),
  CREATED DATE,
  CREATED_BY VARCHAR2(26),
  UPDATED DATE,
  UPDATED_BY VARCHAR2(26),
  DELIVERY_FEE NUMBER(3),
  DELIVERY_TIME NUMBER(4)
)
;
```

```
#INFO Table created
```

```
#INFO Number of rows processed: 11
```

```
#INFO Number of rows in error: 0
```

```
#INFO Last row processed in final committed batch: 11
```

```
SUCCESS: Processed without errors
```



> **load** conference sample_restaurant_data_table.csv

```
SQL> load conference sample_restaurant_data_table.csv
```

```
csv
column_names on
delimiter ,
enclosures ""
double off
encoding UTF8
row_limit off
row_terminator default
skip_rows 0
skip_after_names
```

```
Load data into table DEMO.CONFERENCE
```

```
#INFO Number of rows processed: 11
#INFO Number of rows in error: 0
#INFO Last row processed in final committed batch: 11
SUCCESS: Processed without errors
SQL> |
```




> **help** set loadformat

```
SQL> help set loadformat
```

```
SET LOADFORMAT [ default|csv|delimited|html|insert|json|json-formatted|loader|t2|xml|xls|xlsx] [options...]
```

default	: Load format properties return to default values
csv	: comma separated values
delimited	: (CSV synonym) delimited format, comma separated values by default
html	: UNLOAD only, Hypertext Markup Language
insert	: UNLOAD only, SQL insert statements
json	: UNLOAD only, Java Script Object Notation
json-formatted	: UNLOAD only, "pretty" formatted JSON
loader	: UNLOAD only, Oracle SQLLoader format
t2	: UNLOAD only, T2 Metrics
xml	: UNLOAD only, Extensible Markup Language
xls	: Excel 97- Excel 2003 Workbook format
xlsx	: Excel format for version 2007 or higher

SQLcl Data Pump



Export full schema from DEV [DEMO] [on-premise]

> dp export

```
SQL> dp export
Initiating DATA PUMP
Database Time Zone: VERSION:44 CON_ID:0
Log Location: DATA_PUMP_DIR:ESQL_770.LOG
Starting "DEMO"."ESQL_770":
Processing object type SCHEMA_EXPORT/PRE_SCHEMA/PROCACT_SCHEMA/LOGREP
Processing object type SCHEMA_EXPORT/PACKAGE/PACKAGE_BODY
Processing object type SCHEMA_EXPORT/TABLE/INDEX/STATISTICS/INDEX_STATISTICS
Processing object type SCHEMA_EXPORT/TABLE/STATISTICS/TABLE_STATISTICS
Processing object type SCHEMA_EXPORT/TABLE/TABLE_DATA
Processing object type SCHEMA_EXPORT/PRE_SCHEMA/PROCOBJ/SCHEDULER
Processing object type SCHEMA_EXPORT/SEQUENCE/SEQUENCE
Processing object type SCHEMA_EXPORT/TABLE/COMMENT
Processing object type SCHEMA_EXPORT/TABLE/TABLE
Processing object type SCHEMA_EXPORT/TABLE/IDENTITY_COLUMN
Processing object type SCHEMA_EXPORT/PACKAGE/PACKAGE_SPEC
Processing object type SCHEMA_EXPORT/FUNCTION/FUNCTION
Processing object type SCHEMA_EXPORT/PROCEDURE/PROCEDURE
Processing object type SCHEMA_EXPORT/PACKAGE/COMPILE_PACKAGE/PACKAGE_SPEC/ALTER_PACKAGE_SPEC
Processing object type SCHEMA_EXPORT/FUNCTION/ALTER_FUNCTION
Processing object type SCHEMA_EXPORT/PROCEDURE/ALTER_PROCEDURE
Processing object type SCHEMA_EXPORT/TABLE/INDEX/INDEX
Processing object type SCHEMA_EXPORT/VIEW/VIEW
Processing object type SCHEMA_EXPORT/TABLE/CONSTRAINT/CONSTRAINT
Processing object type SCHEMA_EXPORT/TABLE/TRIGGER
Processing object type SCHEMA_EXPORT/TABLE/CONSTRAINT/REF_CONSTRAINT
. . exported "DEMO"."ESQL_500"          2.4 MB      2218 rows
. . exported "DEMO"."ESQL_498"          2.2 MB      2196 rows
. . exported "DEMO"."ESQL_406"          2.1 MB      2173 rows
```

```
. . exported "DEMO"."EBA_DEMO_DG_PRODUCTS"      9.3 KB      46 rows
. . exported "DEMO"."EBA_DEMO_MD_COMMENTS"      9.2 KB      11 rows
. . exported "DEMO"."EBA_DEMO_DG_EMP"           9 KB       14 rows
. . exported "DEMO"."EBA_DEMO_TREE_EMP"         9 KB       14 rows
. . exported "DEMO"."EBA_UT_CHART_PROJECTS"      8.7 KB      14 rows
. . exported "DEMO"."SAMPLE_RESTAURANT_CATEGORIES" 8.5 KB      10 rows
. . exported "DEMO"."SAMPLE_RESTAURANT_ORDER_ITEMS" 8.1 KB       8 rows
. . exported "DEMO"."EBA_DEMO_MD_STATUS"         8 KB        3 rows
. . exported "DEMO"."EBA_DEMO_SEARCH_CUSTOMERS"  6.9 KB      20 rows
. . exported "DEMO"."SAMPLE_RESTAURANT_ORDERS"   6.6 KB       5 rows
. . exported "DEMO"."EBA_DEMO_IG_DEPT"          6.5 KB       4 rows
. . exported "DEMO"."EBA_DEMO_DG_DEPT"          6.1 KB       4 rows
. . exported "DEMO"."EBA_DEMO_TREE_DEPT"        6.1 KB       4 rows
. . exported "DEMO"."EBA_DEMO_SEARCH_WEB_CREDS"  0 KB        0 rows
. . exported "DEMO"."EBA_VECTOR_AI_HELPER"       0 KB        0 rows
. . exported "DEMO"."EBA_VECTOR_DATATABLE"       0 KB        0 rows
ORA-39173: Encrypted data has been stored unencrypted in dump file set.
Master table "DEMO"."ESQL_770" successfully loaded/unloaded
*****
Dump file set for DEMO.ESQL_770 is:
/u03/dbfs/4B003B659121328EE063BB59000ACD83/data/dpdump/ESQL_770.DMP
Job "DEMO"."ESQL_770" successfully completed at Tue Mar 17 16:47:19 2026 elapsed 0 00:01:15
DataPump Operation Status 'COMPLETED'
```




Export full schema from DEV [DEMO] [OCI]

```
>datapump EXPORT -schemas DEMO -directory DATA_PUMP_DIR -credential
DEMO_OBJ_STORAGE -dumpuri MY_BUCKET_URL/demo_schema.dmp
```

```
. . exported "DEMO"."EBA_DEMO_MD_MILESTONES"          10.8 KB      12 rows
. . exported "DEMO"."EBA_DEMO_MD_TASK_LINKS"           11 KB         7 rows
. . exported "DEMO"."EBA_DEMO_SEARCH_ORDERS"           10.1 KB     105 rows
. . exported "DEMO"."EBA_DEMO_MD_PROJECTS"             10.1 KB         5 rows
. . exported "DEMO"."EBA_DEMO_DG_PRODUCTS"              9.3 KB        46 rows
. . exported "DEMO"."EBA_DEMO_MD_COMMENTS"             9.2 KB        11 rows
. . exported "DEMO"."EBA_DEMO_DG_EMP"                   9 KB          14 rows
. . exported "DEMO"."EBA_DEMO_TREE_EMP"                 9 KB          14 rows
. . exported "DEMO"."EBA_UT_CHART_PROJECTS"             8.7 KB        14 rows
. . exported "DEMO"."SAMPLE_RESTAURANT_CATEGORIES"      8.5 KB         10 rows
. . exported "DEMO"."SAMPLE_RESTAURANT_ORDER_ITEMS"    8.1 KB         8 rows
. . exported "DEMO"."EBA_DEMO_MD_STATUS"                8 KB           3 rows
. . exported "DEMO"."EBA_DEMO_SEARCH_CUSTOMERS"        6.9 KB        20 rows
. . exported "DEMO"."SAMPLE_RESTAURANT_ORDERS"          6.6 KB         5 rows
. . exported "DEMO"."EBA_DEMO_IG_DEPT"                  6.5 KB         4 rows
. . exported "DEMO"."EBA_DEMO_DG_DEPT"                  6.1 KB         4 rows
. . exported "DEMO"."EBA_DEMO_TREE_DEPT"                6.1 KB         4 rows
. . exported "DEMO"."EBA_DEMO_SEARCH_WEB_CREDS"         0 KB           0 rows
. . exported "DEMO"."EBA_VECTOR_AI_HELPER"              0 KB           0 rows
. . exported "DEMO"."EBA_VECTOR_DATATABLE"              0 KB           0 rows
ORA-39173: Encrypted data has been stored unencrypted in dump file set.
Master table "DEMO"."ESQL_772" successfully loaded/unloaded
*****
Dump file set for DEMO.ESQL 772 is:
https://objectstorage[redacted]/demo_schema.dmp
Job "DEMO"."ESQL_772" successfully completed at Tue Mar 17 17:05:51 2026 elapsed 0 00:00:57
DataPump Operation Status 'COMPLETED'
```



Import full schema [DEMO] into TEST

```
> sql -name test_demo
> datapump IMPORT -schemas DEMO -directory DATA_PUMP_DIR -credential
DEMO_OBJ_STORAGE -dumpuri MY_BUCKET_URL/o/demo_schema.dmp
```

```
SQL> DATAPUMP IMPORT -schemas DEMO -directory DATA_PUMP_DIR -credential DEMO_OBJ_STORAGE -dumpuri /bucket-data-pump-demo/o/demo_schema.dmp
Initiating DATA PUMP
Database Time Zone: VERSION:44 CON_ID:0
Log Location: DATA_PUMP_DIR:ISQL_420.LOG
Master table "DEMO"."ISQL_420" successfully loaded/unloaded
Starting "DEMO"."ISQL_420":
Processing object type SCHEMA_EXPORT/PRE_SCHEMA/PROCACT_SCHEMA/LOGREP
Processing object type SCHEMA_EXPORT/PRE_SCHEMA/PROCOBJ/SCHEDULER
ORA-39083: Object type SCHEDULER:PROCOBJ failed to create with error:
ORA-20022: Credential "DEMO"."DEMO_OBJ_STORAGE" already exists
Processing object type SCHEMA_EXPORT/SEQUENCE/SEQUENCE
Processing object type SCHEMA_EXPORT/TABLE/TABLE
Processing object type SCHEMA_EXPORT/TABLE/TABLE_DATA
. . imported "DEMO"."EBA_DEMO_SEARCH_WEB_CREDS"          0 KB          0 rows
. . imported "DEMO"."EBA_VECTOR_DATATABLE"              0 KB          0 rows
. . imported "DEMO"."EBA_DEMO_SEARCH_CUSTOMERS"         6.9 KB         20 rows
. . imported "DEMO"."EBA_DEMO_SEARCH_ORDERS"          10.1 KB        105 rows
. . imported "DEMO"."EBA_DEMO_DG_DEPT"                  6.1 KB          4 rows
. . imported "DEMO"."SAMPLE_RESTAURANT_ITEMS"          12.8 KB         24 rows
. . imported "DEMO"."EBA_UT_DEMO_CARDS"                19.4 KB         18 rows
. . imported "DEMO"."EBA_DEMO_SEARCH_PRODUCTS"         45.1 KB         46 rows
. . imported "DEMO"."EBA_DEMO_MD_TASKS"                19.1 KB         45 rows
. . imported "DEMO"."SAMPLE_RESTAURANT"                12.1 KB         11 rows
. . imported "DEMO"."EBA_DEMO_DG_EMP"                   9 KB           14 rows
. . imported "DEMO"."EBA_DEMO_MD_TASK_TODOS"           11.2 KB          9 rows
. . imported "DEMO"."EBA_DEMO_IG_DEPT"                  6.5 KB          4 rows
. . imported "DEMO"."EBA_DEMO_MD_PROJECTS"             10.1 KB          5 rows
. . imported "DEMO"."EBA_DEMO_DG_CUSTOMERS"             25 KB          392 rows
. . imported "DEMO"."EBA_DEMO_IG_EMP"                  12.6 KB         15 rows
. . imported "DEMO"."EBA_DEMO_TREE_STOCKS"             11.6 KB         11 rows
. . imported "DEMO"."ESQL_770"                         2.6 MB        2240 rows
. . imported "DEMO"."EBA_DEMO_SEARCH_STORES"           11.6 KB         23 rows
. . imported "DEMO"."ESQL_408"                         2.2 MB        2106 rows
```




Import full schema [DEMO] into TEST

```
> sql -name test_demo
> datapump IMPORT -schemas DEMO -directory DATA_PUMP_DIR -credential
DEMO_OBJ_STORAGE -dumpuri MY_BUCKET_URL/o/demo_schema.dmp
```

```
. . imported "DEMO"."SAMPLE_RESTAURANT_ORDERS"          6.6 KB      5 rows
. . imported "DEMO"."SAMPLE_RESTAURANT_ORDER_ITEMS"     8.1 KB      8 rows
. . imported "DEMO"."EBA_DEMO_TREE_DEPT"                6.1 KB      4 rows
. . imported "DEMO"."CONFERENCE"                       13.8 KB     22 rows
. . imported "DEMO"."ESQL_500"                          2.4 MB    2218 rows
. . imported "DEMO"."EBA_DEMO_MD_COMMENTS"              9.2 KB     11 rows
. . imported "DEMO"."SAMPLE_RESTAURANT_NEW"             12.2 KB     11 rows
. . imported "DEMO"."EBA_DEMO_DG_PRODUCTS"              9.3 KB     46 rows
. . imported "DEMO"."EBA_DEMO_MD_MILESTONES"            10.8 KB     12 rows
. . imported "DEMO"."EBA_VECTOR_MOVEDATA"               10 KB      10 rows
. . imported "DEMO"."EBA_DEMO_MD_TEAM_MEMBERS"          11 KB      12 rows
. . imported "DEMO"."SAMPLE_RESTAURANT_CATEGORIES"      8.5 KB     10 rows
. . imported "DEMO"."EBA_DEMO_TREE_PROJ_FILES"           51 KB       4 rows
. . imported "DEMO"."EBA_UT_MAP_AIRPORTS"               20.7 KB     45 rows
Processing object type SCHEMA_EXPORT/TABLE/COMMENT
Processing object type SCHEMA_EXPORT/TABLE/IDENTITY_COLUMN
Processing object type SCHEMA_EXPORT/PACKAGE/PACKAGE_SPEC
Processing object type SCHEMA_EXPORT/FUNCTION/FUNCTION
Processing object type SCHEMA_EXPORT/PROCEDURE/PROCEDURE
Processing object type SCHEMA_EXPORT/PACKAGE/COMPILE_PACKAGE/PACKAGE_SPEC/ALTER_PACKAGE_SPEC
Processing object type SCHEMA_EXPORT/FUNCTION/ALTER_FUNCTION
Processing object type SCHEMA_EXPORT/PROCEDURE/ALTER_PROCEDURE
Processing object type SCHEMA_EXPORT/VIEW/VIEW
Processing object type SCHEMA_EXPORT/PACKAGE/PACKAGE_BODY
Processing object type SCHEMA_EXPORT/TABLE/INDEX/INDEX
Processing object type SCHEMA_EXPORT/TABLE/CONSTRAINT/CONSTRAINT
Processing object type SCHEMA_EXPORT/TABLE/INDEX/STATISTICS/INDEX_STATISTICS
Processing object type SCHEMA_EXPORT/TABLE/CONSTRAINT/REF_CONSTRAINT
Processing object type SCHEMA_EXPORT/TABLE/TRIGGER
Processing object type SCHEMA_EXPORT/TABLE/STATISTICS/TABLE_STATISTICS
Processing object type SCHEMA_EXPORT/POST_SCHEMA/PROCACT_SCHEMA/LBAC_EXP
Job "DEMO"."ISQL_420" completed with 1 error(s) at Tue Mar 17 17:25:42 2026 elapsed 0 00:00:55
DataPump Operation Status 'COMPLETED'
```




Do you want to know more?

> **help** datapump

Oracle Data Pump enables high-speed movement of data and metadata from one database to another.
The datapump command uses the extension of the DBMS_DATAPUMP package.
The subcommands **export** and **import** are available within the **datapump** feature.

Subcommands:

EXPORT

Export schema or list of schemas with the datapump utility using DBMS_DATAPUMP package.

Usage:

export|ex {OPTIONS}

Options:

-compression|-cm <compression> {all|data_only|metadata_only|none}

Default: As specified on set datapump or METADATA_ONLY

Indicates if compression should be done on user data and metadata.

ALL : Compress user data and metadata.

DATA_ONLY : Compress only user data.

METADATA_ONLY : Compress only metadata.

NONE : Do not compress user data or metadata.

-copycloud|-cc <copycloud>

Default: As specified on set datapump or FALSE

[TRUE] : Copy the dumpfile between database directory and Oracle Object Store

On export, copy the dumpfile from the database directory to Oracle Object Store after the datapump job completes.

On import, copy the dumpfile from Oracle Object Store to the database directory before the datapump job starts

OCI PROFILE must be set using OCI command or CLOUDSTORAGE command.

Cloud storage uri must be set using the CLOUDSTORAGE command, SET DATAPUMP command or DATAPUMP command.

SQLcl

APEX [subcommand]



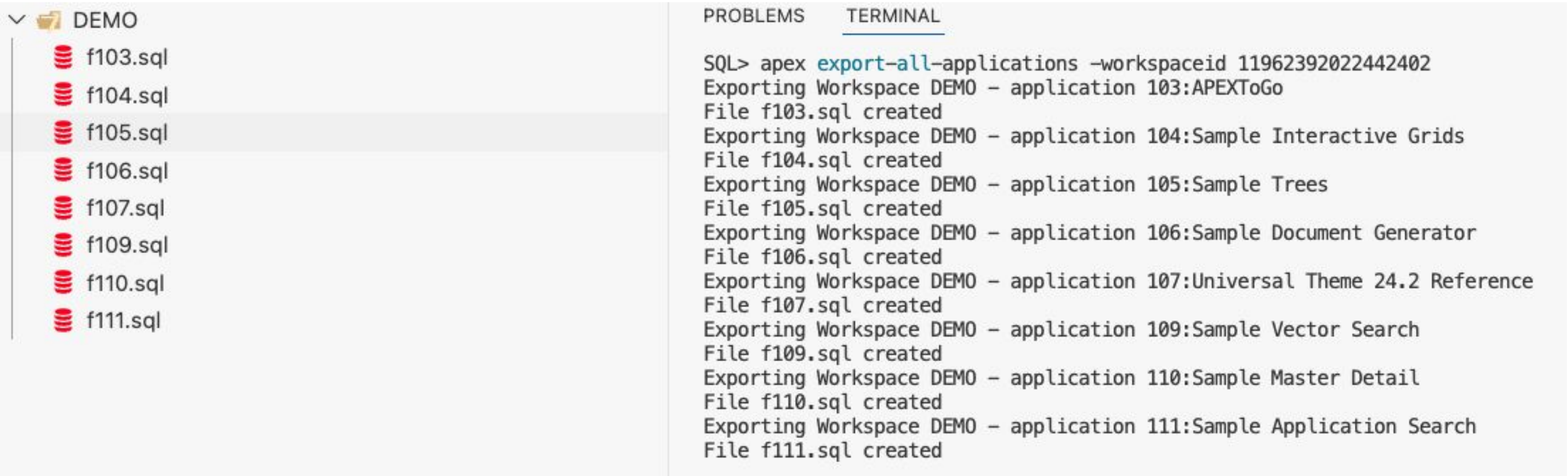
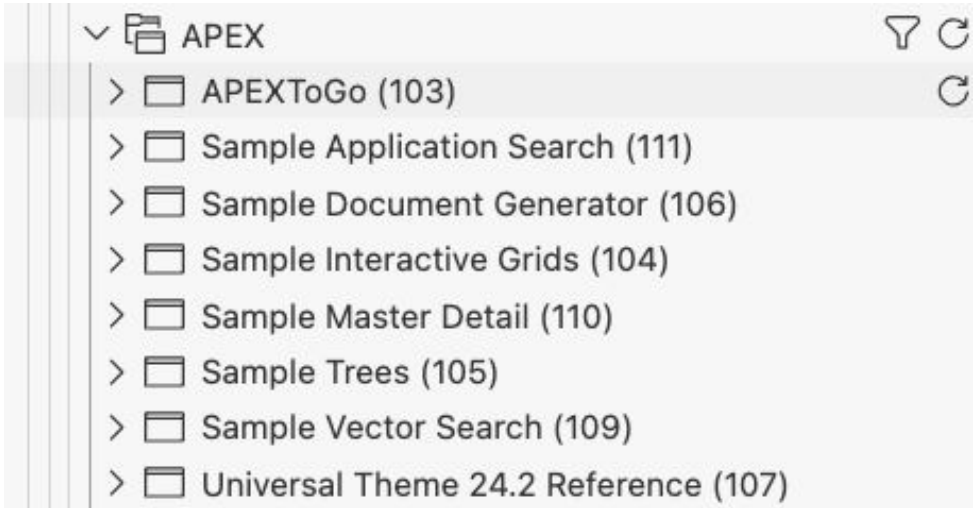
Export APEX Applications using command line

```
> help apex
> apex [subcommand]
> export
> export-all-workspaces (wkspc only, no apps, structure only)
> export-application
> export-components
> list
> log (usage reports)
> version
> (...and few more)
```




Export ALL apps

> apex **export-all-applications** -workspaceid 11962392022442402





Export single app

```
> apex export -applicationid 103 -exptype  
APPLICATION_SOURCE,READABLE_YAML -split -dir apex
```

The screenshot shows an IDE interface. On the left, the file explorer displays a tree structure under the 'apex' folder, with 'f103' expanded to show 'application', 'readable', and 'workspace' folders, along with 'copyright.txt' and 'install.sql' files. The 'workspace' folder is selected. On the right, the terminal window shows the command execution:

```
SQL> apex export -applicationid 103 -exptype APPLICATION_SOURCE,READABLE_YAML -split -dir apex  
Exporting Workspace DEMO - application 103:APEXToGo  
File apex/f103/readable/workspace/app_groups.yaml created  
  
SQL>
```

List - returns all APEX applications

> apex list

```
SQL> apex list
```

WORKSPACE_ID	WORKSPACE	APPLICATION_ID	APPLICATION_NAME	BUILD_STATUS	LAST_UPDATED_ON	LAST_UPDATED_BY
11962392022442402	DEMO	103	APEXToGo	Run and Develop		
11962392022442402	DEMO	104	Sample Interactive Grids	Run and Develop		
11962392022442402	DEMO	105	Sample Trees	Run and Develop		
11962392022442402	DEMO	106	Sample Document Generator	Run and Develop		
11962392022442402	DEMO	107	Universal Theme 24.2 Reference	Run and Develop		
11962392022442402	DEMO	109	Sample Vector Search	Run and Develop		
11962392022442402	DEMO	110	Sample Master Detail	Run and Develop		
11962392022442402	DEMO	111	Sample Application Search	Run and Develop		


```
> apex log [APEX_WORKSPACE_LOG_SUMMARY]
```

 $\langle\langle \text{time} \rangle\rangle$

—

—

[illegible]



▼

📁 APEX

🔍

↺

>

📁 APEXToGo (103)

↺

> 📁 Sample Application Search (111)

> 📁 Sample Document Generator (106)

> 📁 Sample Interactive Grids (104)

> 📁 Sample Master Detail (110)

> 📁 Sample Trees (105)

> 📁 Sample Vector Search (109)

> 📁 Universal Theme 24.2 Reference (107)

> **help** apex export

EXPORT

Export APEX applications and or workspaces.

Usage:

export|ex {OPTIONS}

Options:

-applicationid|-api <applicationid> {INTEGER}
ID for application to be exported

-workspaceid|-woi <workspaceid> {INTEGER}
Workspace ID for which all applications to be exported or the workspace to be exported

-instance|-in
Export all applications

-expworkspace|-exwo
Export workspace identified by -workspaceid or all workspaces if -workspaceid not specified

-expminimal|-exmi
Only export workspace definition, users, and groups

-expfiles|-exfi
Export all workspace files identified by -workspaceid

SQLcl Project (CI/CD)



- “Project ” is an CI/CD extension in SQLcl
- Introduced in 2nd half of 2024
- Contains several sub-commands to help developers manage database & APEX changes
- It uses **Liquibase** & **GIT** in a simple & developer-friendly way



Initialize a new project (GIT repo required)

- > `sql -name dev_demo`
- > `project init -name DEMO -schemas DEMO`

```
DEMO@dev_demo > project init -name DEMO -schemas DEMO
```

PROJECT DETAILS

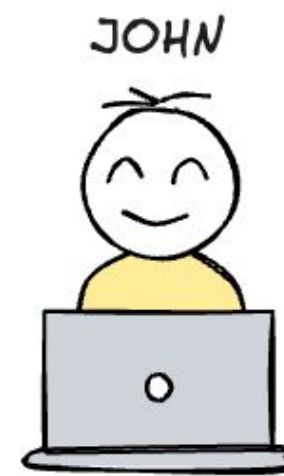
```
-----  
Project name:    DEMO  
Schema(s):      DEMO  
Directory:      /Users/rg/DEMO  
Connection name: dev_demo  
Project root:    DEMO  
Your project has been successfully created  
DEMO@dev_demo >
```

A file explorer view showing the contents of the DEMO project. The project folder is expanded, revealing subfolders .dbtools, dist, and src, and files .gitignore and README.md. The .gitignore and README.md files are marked with a 'U' icon, indicating they are untracked.

```
DEMO  
├── .dbtools  
├── dist  
├── src  
├── .gitignore  
└── README.md
```




Make some Database Objects & APEX changes





Export ALL database & APEX objects

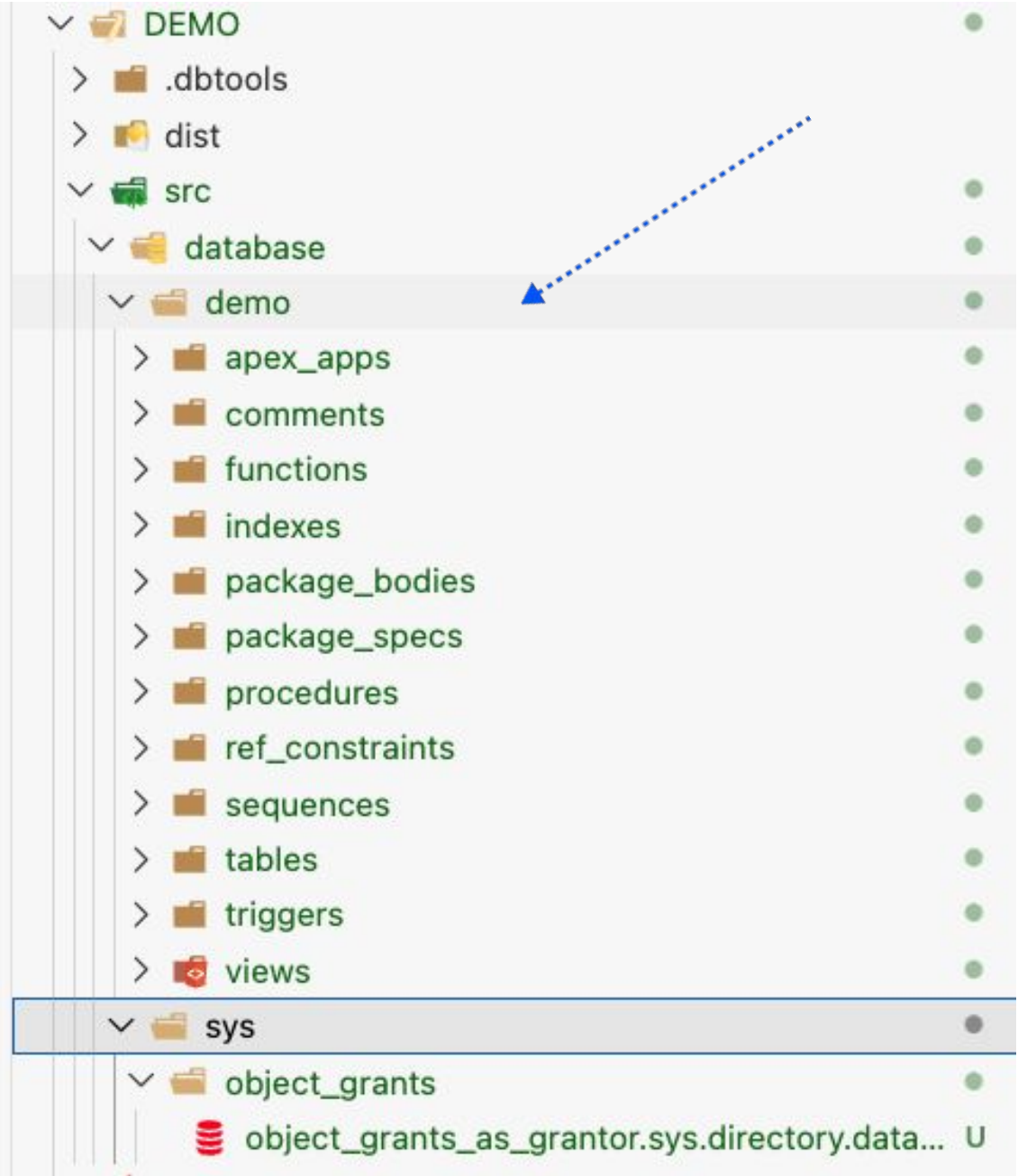
COUNT(*)	OBJECT_TYPE
142	INDEX
51	TABLE
30	LOB
1	CREDENTIAL
23	TRIGGER
6	PACKAGE
7	VIEW
2	FUNCTION
6	PACKAGE BODY
14	SEQUENCE
2	PROCEDURE

▼	📁 APEX	🔍 ↻
>	📁 APEXToGo (103)	↻
>	📁 Sample Application Search (111)	
>	📁 Sample Document Generator (106)	
>	📁 Sample Interactive Grids (104)	
>	📁 Sample Master Detail (110)	
>	📁 Sample Trees (105)	
>	📁 Sample Vector Search (109)	
>	📁 Universal Theme 24.2 Reference (107)	

Export ALL database & APEX objects

> project export

```
DEMO@dev_demo > project export
The current connection DEMO will
*** TABLES ***
*** OBJECT_GRANTS ***
*** APEX_APPLICATIONS ***
*** VIEWS ***
*** FUNCTIONS ***
*** PACKAGE_SPECS ***
*** PACKAGE_BODIES ***
*** PROCEDURES ***
*** SEQUENCES ***
*** INDEXES ***
*** TRIGGERS ***
*** REF_CONSTRAINTS ***
Exporting Workspace DEMO - appli
Exporting Workspace DEMO - appli
Exporting Workspace DEMO - appli
Exporting Workspace DEMO - appli
Exporting Workspace DEMO - appli
Exporting Workspace DEMO - appli
Exporting Workspace DEMO - appli
Exporting Workspace DEMO - appli
Exporting Workspace DEMO - appli
Exporting Workspace DEMO - appli
-----
APEX_APPLICATION      8
COMMENT               5
FUNCTION              2
GRANT                 2
INDEX                61
PACKAGE_BODY          6
PACKAGE_SPEC          6
PROCEDURE             2
REF_CONSTRAINT       30
SEQUENCE              3
TABLE                51
TRIGGER              23
VIEW                  7
-----
Exported 206 objects
Elapsed 208 sec
```





Project Stage

- Compares the current branch (**dev**) with the target branch (**test**) to create Liquibase changelogs that apply the necessary database updates.

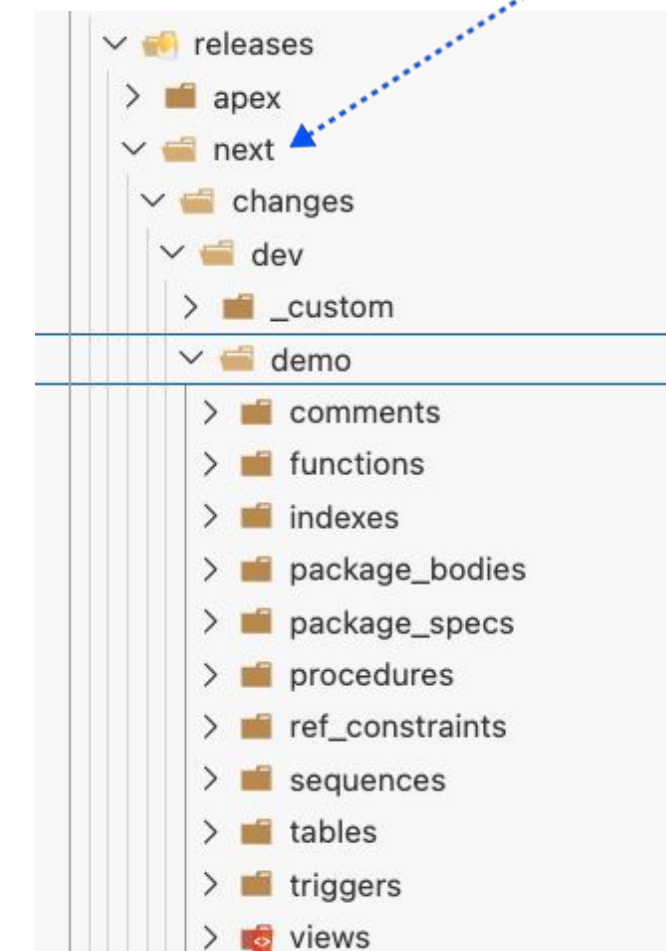
> project **stage** -branch-name test

```
DEMO@dev_demo > project stage -branch-name test
The current connection DEMO will be used for all operations
```

Stage is Comparing:

```
Old Branch    refs/heads/test
New Branch    refs/heads/dev
```

```
Stage processing completed, please review and commit your changes to repository
```



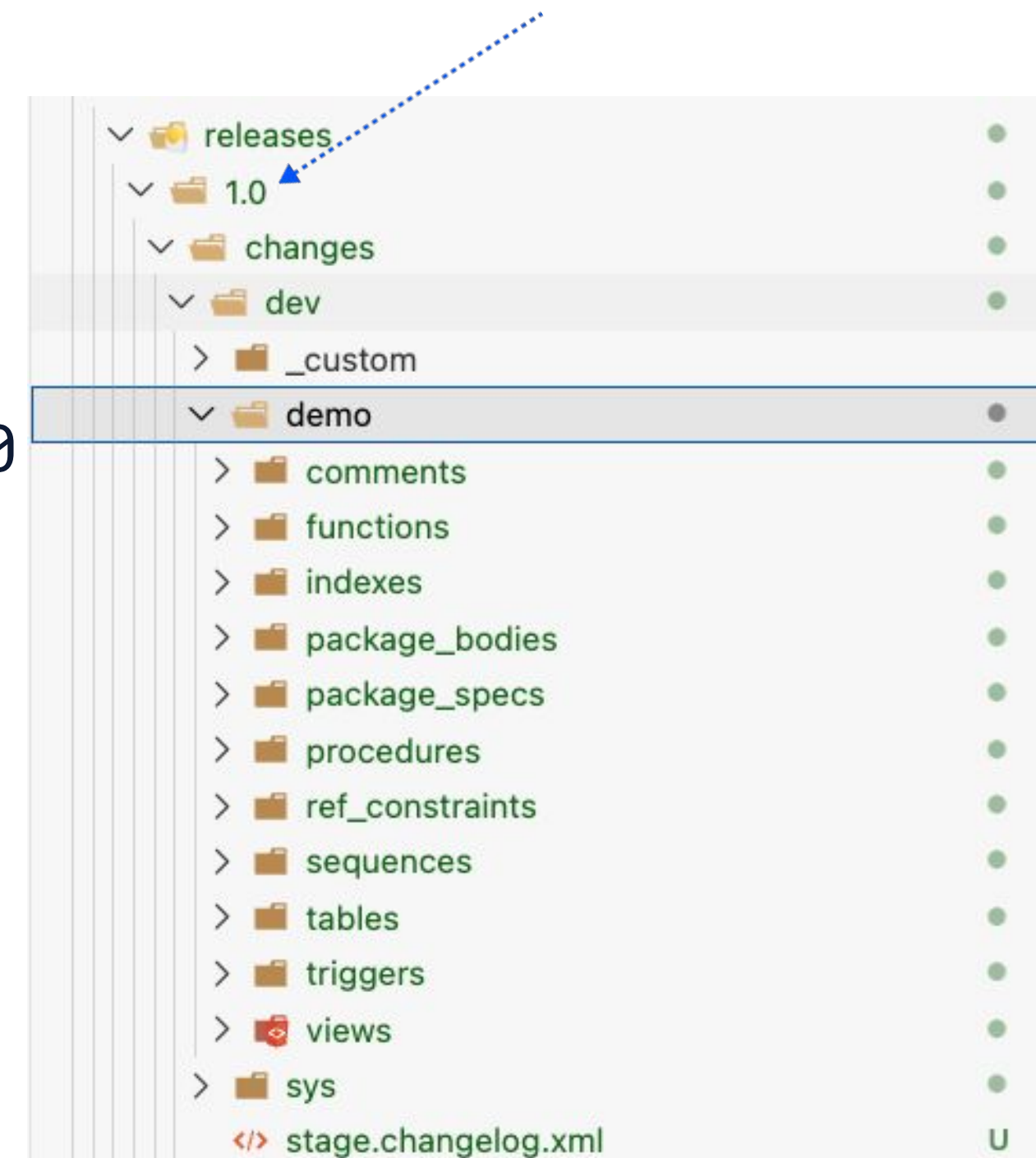


Project Release

- Transitions current work set into a released state and initiates a fresh set of work
- Once the code is moved into a release, it must not be altered.

> project **release** -version 1.0

```
DEMO@dev_demo > project release -version 1.0
Process completed successfully
DEMO@dev_demo > |
```





Liquibase syntax and files are automatically created:

eba_demo_dg_orders.sql U X

DEMO > dist > releases > 1.0 > changes > dev > demo > tables > eba_demo_dg_orders.sql > ...

```
1  -- liquibase formatted sql
2  -- changeset DEMO:1773775410323 stripComments:false logicalFilePath:dev/demo/tables/eba_demo_dg_orders.sql
3  -- sqlcl_snapshot src/database/demo/tables/eba_demo_dg_orders.sql:null:dc1ca737d177f7fef60aaa0dffff9f9bf1b8458e:create
4
5  create table demo.eba_demo_dg_orders (
6      order_id      number generated by default on null as identity not null enable,
7      order_datetime timestamp(6) not null enable,
8      customer_id    number not null enable,
9      order_status   varchar2(10 byte) not null enable
10 );
```

stage.changelog.xml U X

DEMO > dist > releases > 1.0 > changes > dev > stage.changelog.xml

```
1  <?xml version="1.0" encoding="UTF-8"?>
2  <databaseChangeLog xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
3      xmlns="http://www.liquibase.org/xml/ns/dbchangelog"
4      xsi:schemaLocation="http://www.liquibase.org/xml/ns/dbchangelog
5          http://www.liquibase.org/xml/ns/dbchangelog/dbchangelog-4.3.xsd">
6      <include file="sys/object_grants/object_grants_as_grantor.sys.directory.data_pump_dir.sql" relativeToChangelogFile="true"/>
7      <include file="demo/sequences/eba_demo_ig_dept_seq.sql" relativeToChangelogFile="true"/>
8      <include file="demo/sequences/eba_demo_ig_emp_seq.sql" relativeToChangelogFile="true"/>
9      <include file="demo/sequences/eba_demo_ig_people_seq.sql" relativeToChangelogFile="true"/>
10     <include file="demo/functions/varchar2_to_blob.sql" relativeToChangelogFile="true"/>
11     <include file="demo/package_specs/eba_demo_ig_text_pkg.sql" relativeToChangelogFile="true"/>
12     <include file="demo/package_specs/eba_demo_md_data_pkg.sql" relativeToChangelogFile="true"/>
13     <include file="demo/package_specs/eba_demo_search_data_pkg.sql" relativeToChangelogFile="true"/>
14     <include file="demo/package_specs/eba_demo_search_text_pkg.sql" relativeToChangelogFile="true"/>
15     <include file="demo/package_specs/eba_sample_vector_search_pkg.sql" relativeToChangelogFile="true"/>
16     <include file="demo/tables/eba_demo_dg_dept.sql" relativeToChangelogFile="true"/>
17     <include file="demo/tables/eba_demo_dg_emp.sql" relativeToChangelogFile="true"/>
18     <include file="demo/tables/eba_demo_ig_dept.sql" relativeToChangelogFile="true"/>
```

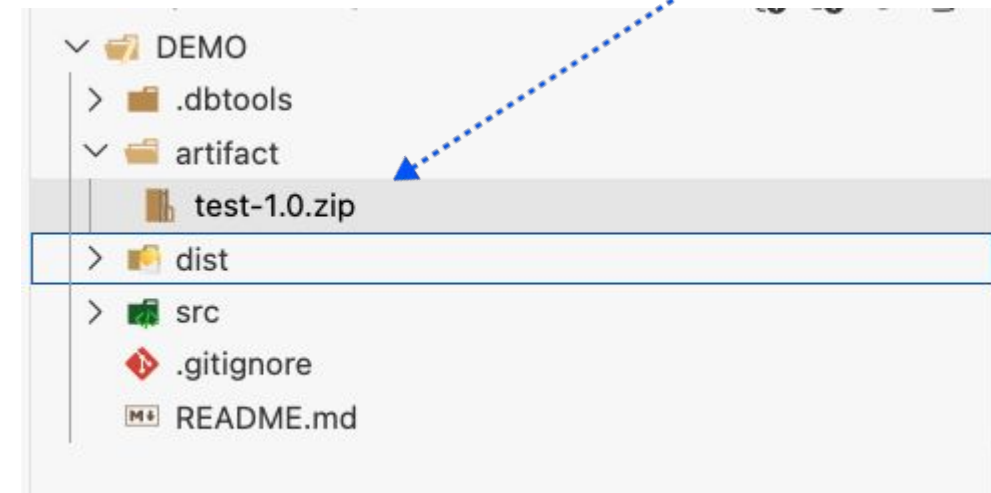


Project Gen-Artifact

- Generates an artifact representing the current state of your project.

> project **gen-artifact** -name test -version 1.0

```
DEMO@dev_demo > project gen-artifact -name test -version 1.0  
Your artifact has been generated test-1.0.zip  
DEMO@dev_demo >
```





Project Deploy

- Deploys the artifact on a target database (TEST)

> project **deploy** -file artifact/test-1.0.zip

```
ADMIN@test_admin > project deploy -file artifact/test-1.0.zip  
Starting the migration...
```



Commands summary

❑ Initialize a new project

```
project init -name bombshell -schemas HR,CO
```

❑ Export all DB objects

```
project export
```

❑ Export some objects

```
project export -objects HR.COUNTRIES,APEX.101
```

❑ Add custom file/script

```
project stage add-custom -file-name my_file
```

❑ Stage (gen. changelogs/changesets)

```
project stage -branch-name test
```

❑ Release

```
project release -version 1.1
```

❑ Generate artifact (zip)

```
project gen-artifact -name test -version 1.1
```

❑ Deploy

```
project deploy -file artifact/test-1.1.zip
```

SQLcl Liquibase



Project (CI/CD) vs. Liquibase

- Liquibase (Open Source) commands are available in SQLcl

```
DEMO@dev_demo > liquibase version
--Starting Liquibase at 2026-03-17T21:00:22.939926 using Java 25.0.2 (version 4.33.0 #0 built at 2025-12-09 17:47+0000)
Liquibase version: 4.33.0 #0 built at 2025-12-09 17:47+0000
Extension Version: 25.4.0.0

Operation completed successfully.
```

> liquibase | lb [command]

> **help** liquibase





- Use it as you would use standalone Liquibase
- Enhanced to support most of Oracle Database Objects
- Bonus: APEX files support (for free)





SQLcl Liquibase at Cloud World 2024 by Jeff Smith & Me





- ```
> sql -name dev_demo
> liquibase generate-schema -split -sql
```

Generates current state of the database into files

```
DEMO@dev_demo > liquibase generate-schema -split -sql
--Starting Liquibase at 2026-03-18T20:13:07.913016 using Java

Export Flags Used:

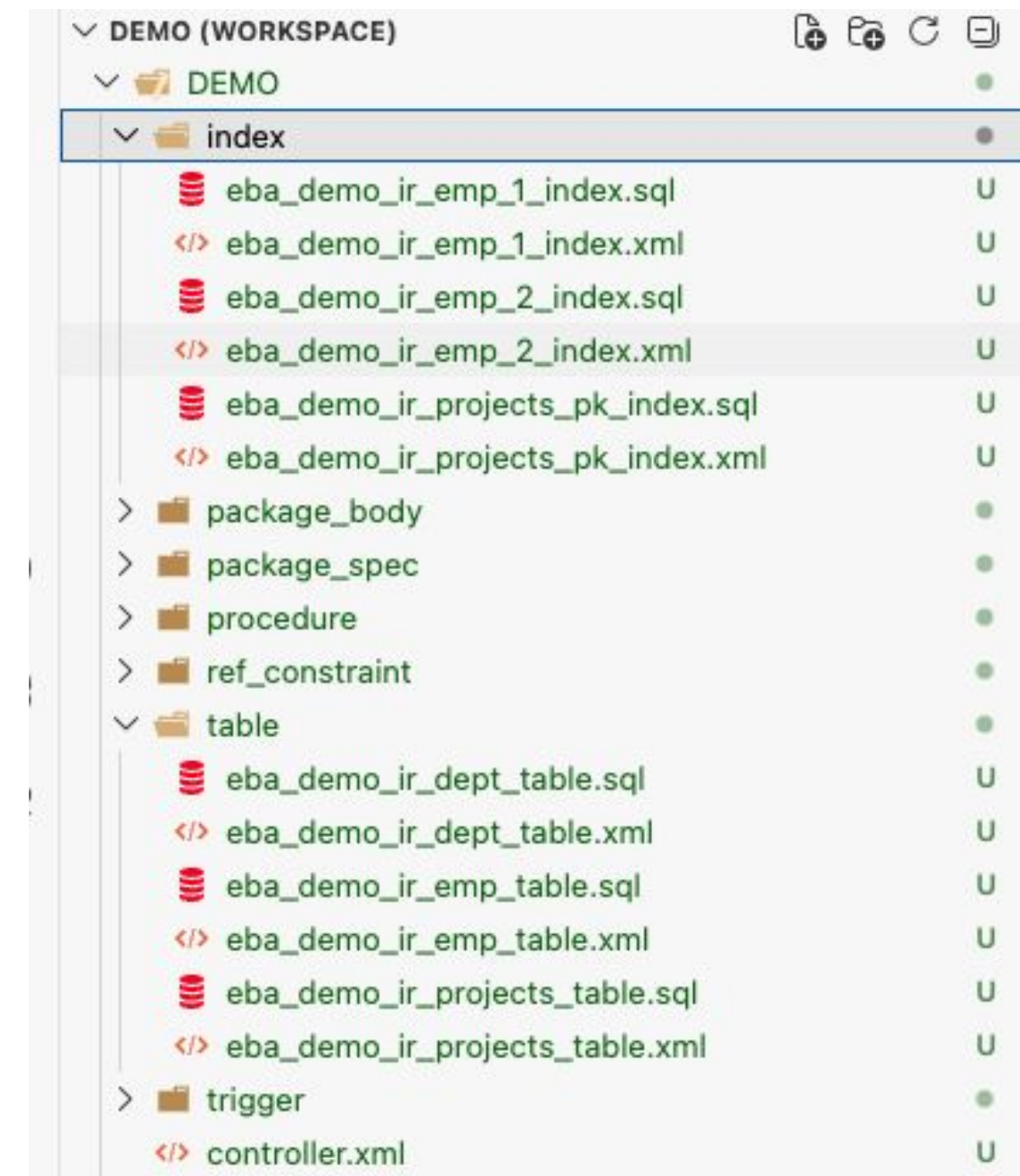
Export Grants false
Export Synonyms false

[Method loadCaptureTable]:

[Method loadCaptureTable]: 1ms
[Load Type DIRECTORY]: 2s 200ms
[Load Type SEQUENCE]: 2s 200ms
[Load Type TYPE_SPEC]: 109ms
[Load Type TYPE_BODY]: 67ms
[Load Type MATERIALIZED_VIEW]: 94ms
[Load Type MATERIALIZED_VIEW_LOG]: 105ms
[Load Type DIMENSION]: 324ms
[Load Type REF_CONSTRAINT]: 407ms
[Load Type CLUSTER]: 15ms
[Load Type FUNCTION]: 390ms
[Load Type PACKAGE_SPEC]: 195ms
[Load Type PROCEDURE]: 39ms
[Load Type VIEW]: 240ms
[Load Type SYNONYM]: 68ms
[Load Type DB_LINK]: 30ms
[Load Type JOB]: 335ms
[Load Type PACKAGE_BODY]: 394ms
[Load Type TRIGGER]: 51ms
[Load Type INDEX]: 3s 546ms
[Load Type TABLE]: 10s 904ms
[Method sortCaptureTable]: 283ms
[Method writeChangeLogs]: 59s 506ms
[Method writeController]: 150ms
[Total Time (Wall Clock)]: 1m 1s 377ms
Changelog created and written out to file controller.xml

Operation completed successfully.
```

```
DEMO@dev_demo > |
```







## Apply changes to another environment (TEST)

- > sql -name **test\_demo**
- > liquibase **update** -changelog-file controller.xml

```
controller.xml U X
DEMO > controller.xml
1 <?xml version="1.0" encoding="UTF-8"?>
2 <databaseChangeLog
3 xmlns="http://www.liquibase.org/xml/ns/dbchangelog"
4 xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
5 xmlns:ext="http://www.oracle.com/xml/ns/dbchangelog-ext"
6 xsi:schemaLocation="http://www.liquibase.org/xml/ns/dbchangelog
7 http://www.liquibase.org/xml/ns/dbchangelog/dbchangelog-latest.xsd">
8 <include file="table/eba_demo_ir_dept_table.xml"/>
9 <include file="table/eba_demo_ir_emp_table.xml"/>
10 <include file="table/eba_demo_ir_projects_table.xml"/>
11 <include file="ref_constraint/sys_c0030965_ref_constraint.xml"/>
12 <include file="package_spec/eba_demo_ir_pkg_package_spec.xml"/>
13 <include file="procedure/eba_demo_ir_data_procedure.xml"/>
14 <include file="index/eba_demo_ir_emp_1_index.xml"/>
15 <include file="index/eba_demo_ir_emp_2_index.xml"/>
16 <include file="index/eba_demo_ir_projects_pk_index.xml"/>
17 <include file="trigger/biu_eba_demo_ir_projects_trigger.xml"/>
18 <include file="package_body/eba_demo_ir_pkg_package_body.xml"/>
19 </databaseChangeLog>
```

```
PROBLEMS TERMINAL
DEMO@test_demo > liquibase update -changelog-file controller.xml
--Starting Liquibase at 2026-03-18T20:25:48.759223 using Java 25.0.2 (version 4.33.0 #0 built at 2025-12-09 17:47+0000)
Running Changeset: table/eba_demo_ir_dept_table.xml::5a1d44dbc39d1b71208a40d44532e604c12a62f2::(DEMO)-Generated
Table "EBA_DEMO_IR_DEPT" created.
Running Changeset: table/eba_demo_ir_emp_table.xml::4b41dd083ed44f7d1b26bb2ea0353af6fabe9843::(DEMO)-Generated
[]
```



## SQLcl Liquibase is super smart

- It exactly knows if change should be deployed or not
- Example:
  - I added a new column NEW\_COL1 into **EBA\_DEMO\_IR\_DEPT** table at DEV
  - Meanwhile someone added same column directly at TEST (ignoring CI/CD process)



# SQLcl Liquibase is super smart

```
DEMO@test_demo >
```





## Generates current state of the chosen object

```
> liquibase generate-db-object -obn -eba_demo_ir_dept -obt table -sql
```

```
PROBLEMS TERMINAL
DEMO@dev_demo > liquibase generate-db-object -object-name eba_demo_ir_dept -object-type table -sql
```



# Generate APEX changeset

> liquibase generate-apex-object -applicationid 103

```
DEMO@dev_demo > liquibase generate-apex-object -applicationid 103
```

```
<?xml version="1.0" encoding="UTF-8"?>
<databaseChangeLog
 xmlns="http://www.liquibase.org/xml/ns/dbchangelog"
 xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
 xmlns:ext="http://www.oracle.com/xml/ns/dbchangelog-ext"
 xsi:schemaLocation="http://www.liquibase.org/xml/ns/dbchangelog
 http://www.liquibase.org/xml/ns/dbchangelog/dbchangelog-latest.xsd">
 <changeSet id="INSTALL_103" author="DEMO">
 <n0:runApexScript objectName="install" objectType="SCRIPT" ownerName="DEMO" sourceType="STRING" >
 <n0:source><![CDATA[
declare
 -- sqlcl version = 25.4.0.0
 -- override_schema = ${lb.apex.schema}
 -- override_alias = ${lb.apex.alias}
 -- override_workspace = ${lb.apex.workspace}
 -- override_app_id = ${lb.apex.appId}

 l_app_id apex_applications.application_id%type := q'[103]';
 l_workspace apex_workspaces.workspace%type := q'[DEMO]';

 l_override_workspace apex_workspaces.workspace%type := q'[${lb.apex.workspace}]';
 l_override_schema apex_workspaces.schemas.schema%type := q'[${lb.apex.schema}]';
 l_override_alias apex_applications.alias%type := q'[${lb.apex.alias}]';
 l_override_app_id apex_applications.application_id%type := q'[${lb.apex.appId}]';
 l_generate_offset boolean := false;

begin
 apex_application_install.clear_all();
 -- set workspace
 if (l_override_workspace is not null and l_override_workspace != l_workspace) then
 apex_application_install.set_workspace(l_override_workspace);
 else
 apex_application_install.set_workspace(l_workspace);
 end if;
 commit;
 -- set app id
 if (l_override_app_id is not null and l_override_app_id != l_app_id) then
 apex_application_install.set_application_id(l_override_app_id);
 l_generate_offset := true;
 else
 apex_application_install.set_application_id(l_app_id);
 end if;
 -- ensure supporting objects get installed
 apex_application_install.set_auto_install_sup_obj(p_auto_install_sup_obj => true);
 -- set schema if not same
 if (l_override_schema is not null) then
 apex_application_install.set_schema(l_override_schema);
 end if;
 -- set alias
 if (l_override_alias is not null) then
 apex_application_install.set_application_alias(l_override_alias);
 end if;
end;
]]]>
 </n0:runApexScript>
 </changeSet>
</databaseChangeLog>
```



## How Liquibase works?

- Liquibase tracks which database changes has been applied to which environment
- Keeps your database structure consistent across dev, test,prod etc.
- Changes are tracked in DATABASECHANGELOG, and executed SQL can be found in DATABASECHANGELOG\_ACTIONS

```
DEMO@test_demo > select id, filename, dateexecuted,comments, md5sum from databasechangelog;
```





# SQLcl Codescan



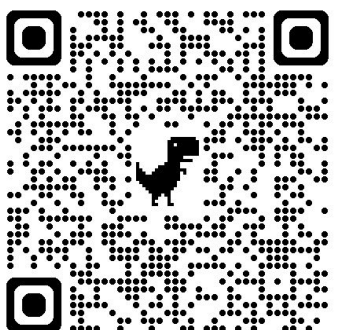
# What is CODESCAN?

- Checks quality of written PL/SQL and SQL
- Follows the rules from Trivadis PL/SQL & SQL Coding Guidelines
- Identifies performance issues stated in Oracle documentation

[SQL Performance Issues](#)

[Trivadis PL/SQL & SQL Coding Guidelines](#)

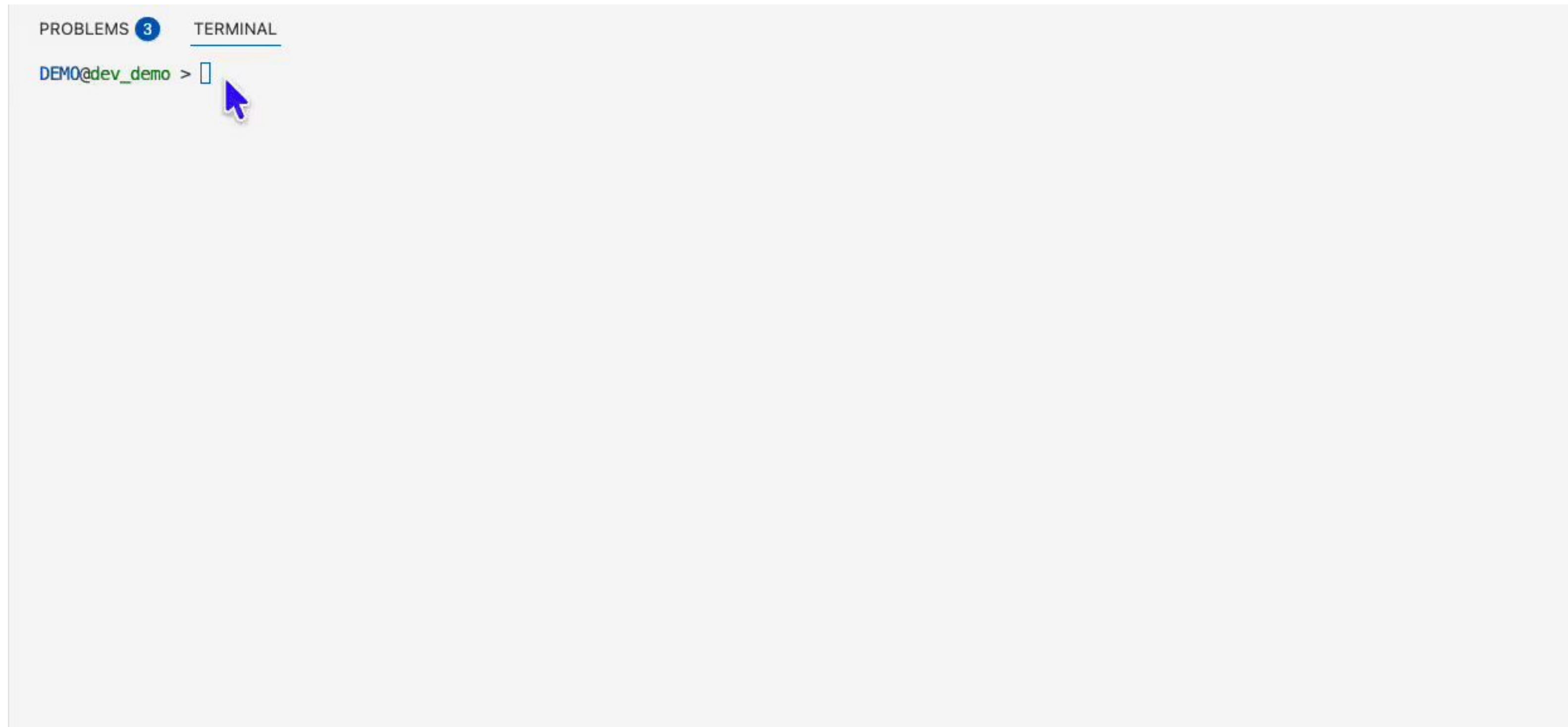
My blog about CODESCAN [here](#) ->





# LIVE quality scan

```
> set codescan on [default off]
```





LIVE scan of  
executed scripts



my\_script.sql 9, U X

DEMO > my\_script.sql > ...

```
1 --G-1030 "Avoid defining variables that are not used"
2 DECLARE
3 l_dummy NUMBER := 1;
4 BEGIN
5 NULL;
6 END;
7 /
8 --G-3145 "Avoid using SELECT * directly from a table or view"
9 SELECT *
10 FROM eba_demo_ir_emp
11 WHERE ename = 'KING';
12
13 --SQL best practice warning (1,19): G-7410: Avoid standalone functions put your functions in packages
14 --SQL best practice warning (1,60): G-7160: Always explicitly state parameter mode
15 CREATE OR REPLACE FUNCTION perf_sample_get_employee_status(p_empno VARCHAR2)
16 RETURN VARCHAR2 IS
17 BEGIN
18 RETURN 'STATUS_' || p_empno;
19 END perf_sample_get_employee_status;
20 /
```

TERMINAL

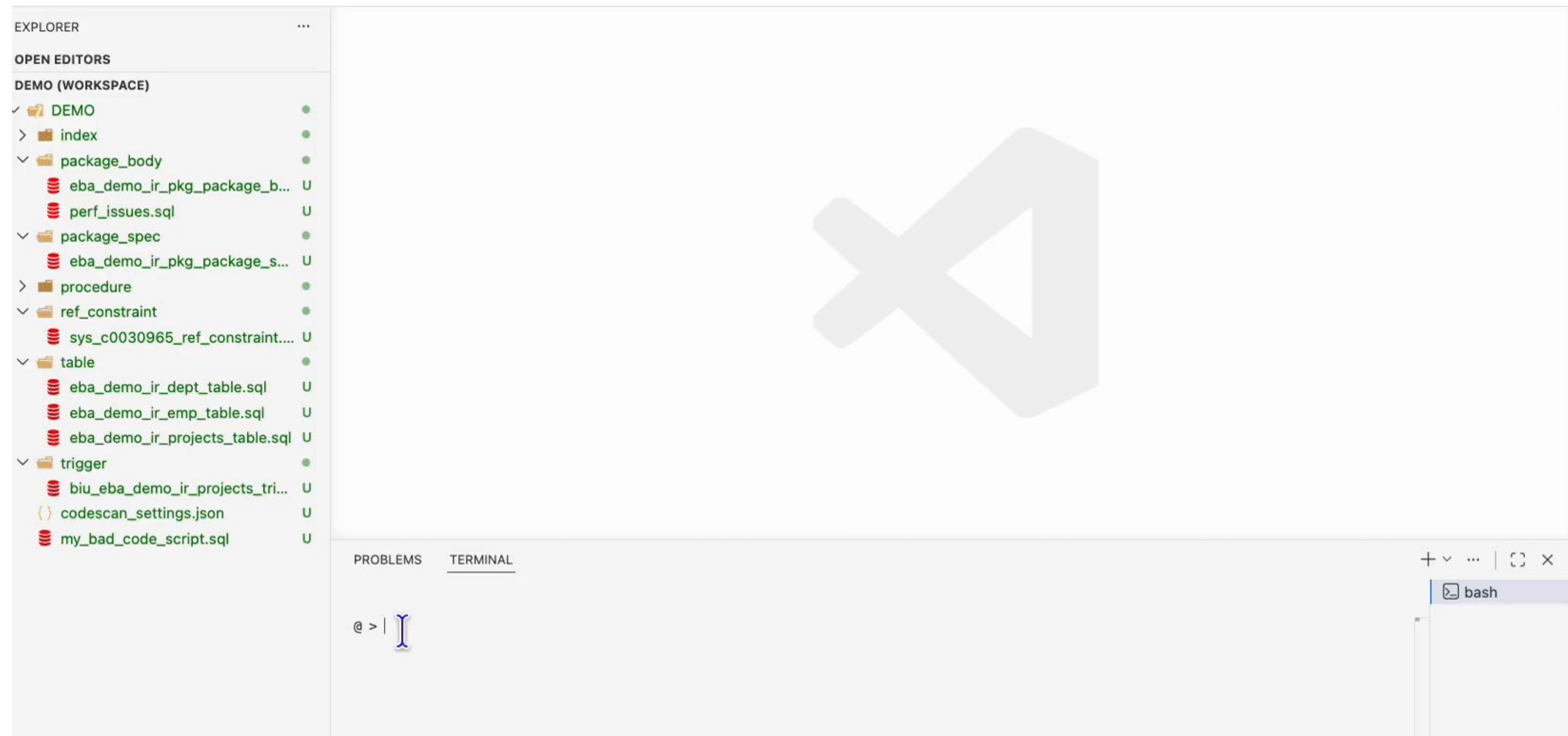
DEMO@dev\_demo > |

00:00:00.660



# Scanning multiple files (**connection not required!**)

> **codescan** -format json -output my\_default\_scan.json







# Ignore rules you don't like / don't agree with

```

1 [
2 {"file": "/Users/rg/DEMO/my_bad_code_script.sql", "issues": [
3 {"line": 2, "col": 2, "ruleNo": "G-1030", "msg": "Avoid defining variables that are not used"},
4 {"line": 8, "col": 7, "ruleNo": "G-3145", "msg": "Avoid using SELECT * directly from a table or view"},
5 {"line": 14, "col": 18, "ruleNo": "G-7410", "msg": "Avoid standalone functions put your functions in packages"},
6 {"line": 14, "col": 59, "ruleNo": "G-7160", "msg": "Always explicitly state parameter mode"},
7 {"line": 24, "col": 1, "ruleNo": "G-3130", "msg": "Try to use ANSI SQL-92 join syntax"},
8 {"line": 25, "col": 18, "ruleNo": "PSR-203", "msg": "An inline view or table detected that was OUTER-joined on optional side of the join"},
9 {"line": 34, "col": 1, "ruleNo": "G-5060", "msg": "Avoid unhandled exceptions"},
10 {"line": 36, "col": 9, "ruleNo": "PSR-102", "msg": "Scalar subqueries are subqueries in SELECT list. They must return exactly one value"},
11 {"line": 42, "col": 1, "ruleNo": "G-5010", "msg": "Try to use a error/logging framework for your application"},
12 {"line": 42, "col": 1, "ruleNo": "G-7510", "msg": "Always prefix Oracle supplied packages with owner schema name"}
13]},
14 {"file": "/Users/rg/DEMO/trigger/biu_eba_demo_ir_projects_trigger.sql", "issues": [
15 {"line": 2, "col": 7, "ruleNo": "G-7730", "msg": "Avoid multiple DML events per trigger"},
16 {"line": 6, "col": 12, "ruleNo": "G-9030", "msg": "Try to define a default value on conversion errors"},
17 {"line": 6, "col": 69, "ruleNo": "G-5060", "msg": "Avoid unhandled exceptions"},
18 {"line": 14, "col": 32, "ruleNo": "G-5050", "msg": "Avoid use of the RAISE_APPLICATION_ERROR built-in procedure with a hard-coded 20nnn"}
19]},
20 {"file": "/Users/rg/DEMO/package_body/eba_demo_ir_pkg_package_body.sql", "issues": [
21 {"line": 14, "col": 34, "ruleNo": "G-2160", "msg": "Avoid initializing variables using functions in the declaration section"},
22 {"line": 21, "col": 41, "ruleNo": "G-1050", "msg": "Avoid using literals in your code"},
23 {"line": 22, "col": 41, "ruleNo": "G-1050", "msg": "Avoid using literals in your code"}
24]}

```

```

1 {
2 "ignore": [
3 "G-3145",
4 "G-3130",
5 "G-1050"
6]
7 }

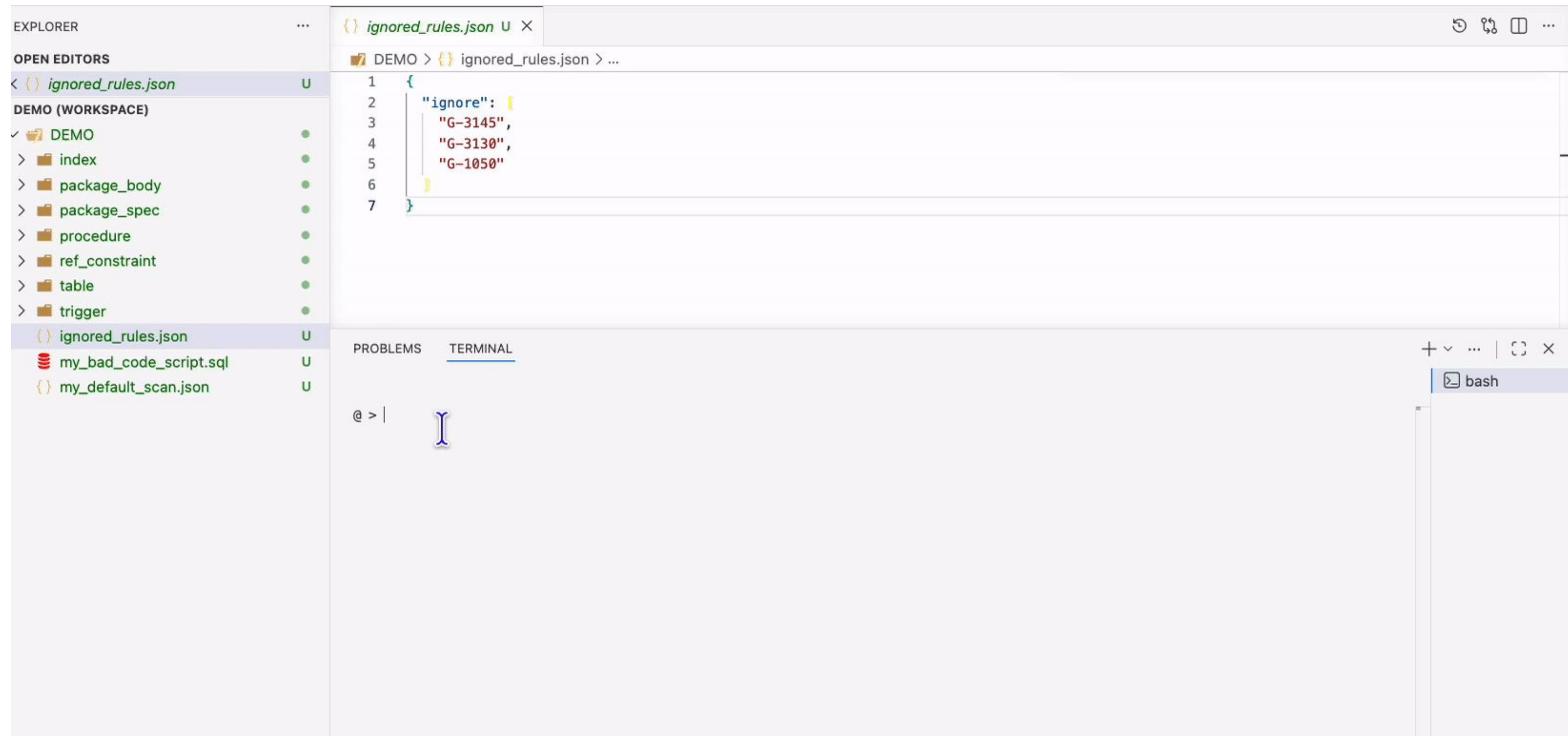
```

\* Ignore doesn't work with Performance Rules (PSR-XXX)



# Ignore rules you don't like / don't agree with

```
> codescan -format json -output custom_scan.json -settings ignored_rules.json
```





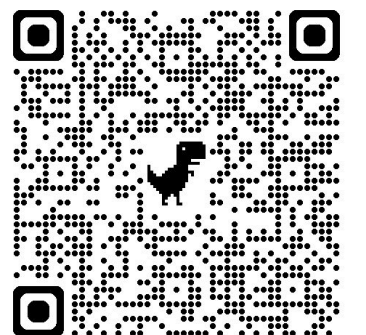
SQLcl  
MCP AI



# What is MCP?

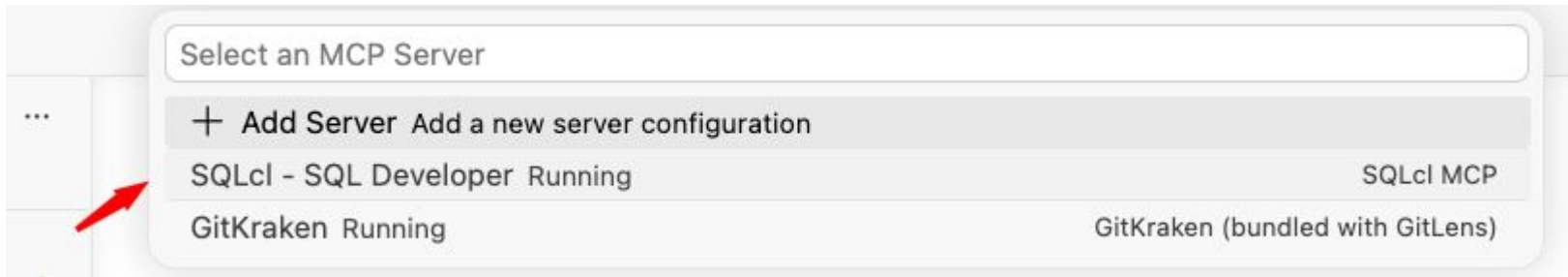
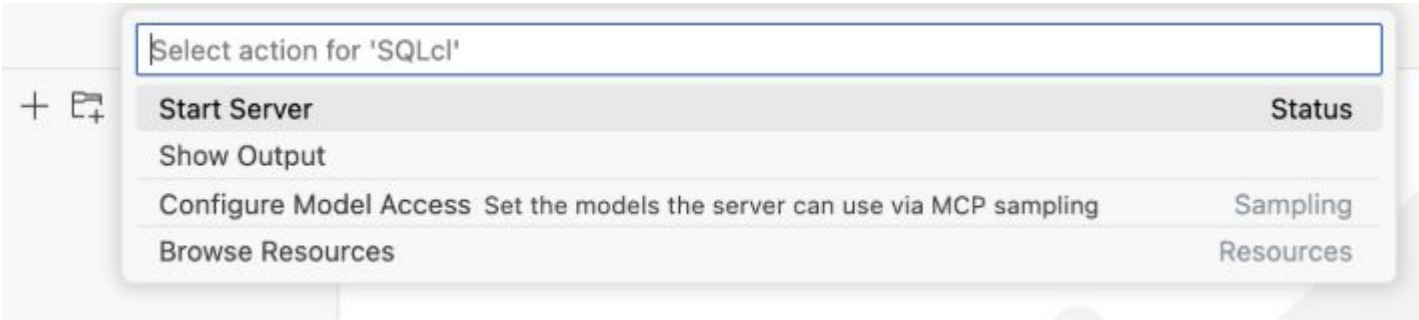
- **Model Context Protocol** is a standard that allows AI models to connect with external data e.g databases

My intro blog about MCP [here](#) ->

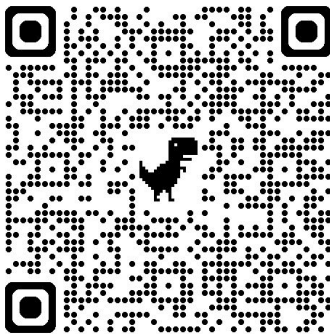


# How to use it in SQLcl ?

- Easiest way via VS Code



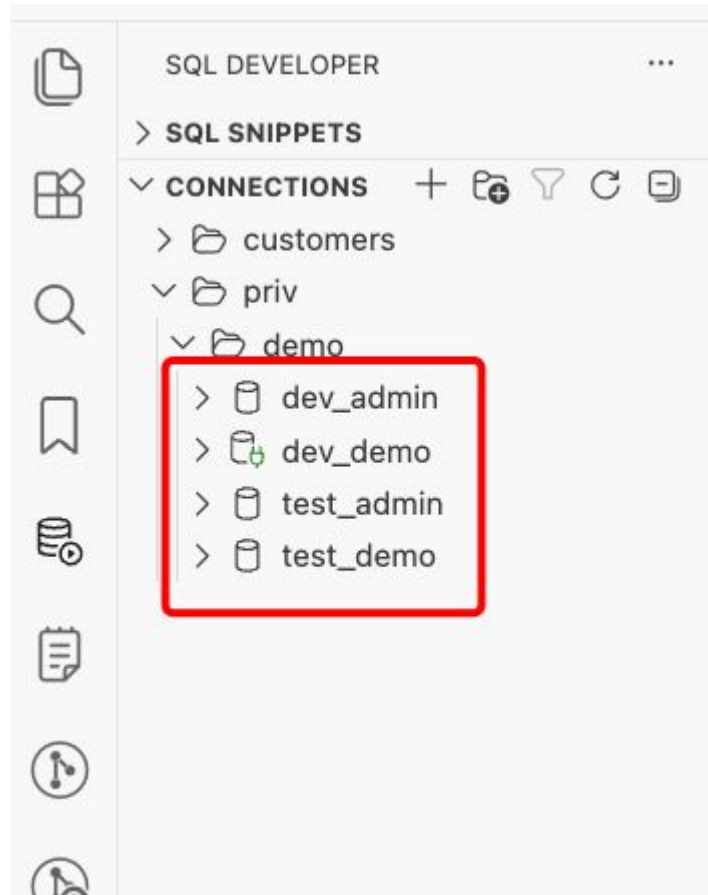
My intro blog about MCP [here](#) ->





## How to use it in SQLcl ?

- Define connection in SQLcl
- Ask AI to use this connection







# AI has access to everything your schema user sees

```
List all tables I have in my schema
```

```
Ran run-sql - SQLcl - SQL Developer (MCP Server) >
```

```
Ran run-sql - SQLcl - SQL Developer (MCP Server)
```

- Connected as DEMO and the schema contains the tables  
table list (there's no file link; only SQL results available): DATABASECHANGELOG\_EXPORT,  
DBTOOLS\$MCP\_LOG, EBA\_DEMO\_IR\_DEPT, EBA\_DEMO\_IR\_EMP, EBA\_DEMO\_IR\_PROJECTS.

🔄 📄 👍 👎 GPT-5.1-Codex-Mini (Preview) • 0.33x



# AI has access to everything your schema user sees

```
CONNECTING TO DEV_DEMO ENVIRONMENT
```

Restore Checkpoint · 🔗

How many tables I have and how much records are there?

Ran run-sql – SQLcl - SQL Developer (MCP Server)

Ran run-sql – SQLcl - SQL Developer (MCP Server)

- There are 5 user tables in dev\_demo according to user\_tables.
- Row counts (up-to-date at query time):
  - EBA\_DEMO\_IR\_DEPT: 4
  - EBA\_DEMO\_IR\_EMP: 15
  - EBA\_DEMO\_IR\_PROJECTS: 73
  - DBTOOLS\$MCP\_LOG: 29
  - DATABASECHANGELOG\_EXPORT: 11

Let me know if you'd like totals by schema, a "biggest tables" ranking, or a periodic refresh of these counts.



# AI has access to everything your schema user sees

Prepare a query grouping employees by department name and show me output as a table

Planned simultaneous table schema reads

Ran `run-sql` – SQLcl - SQL Developer (MCP Server)

Grouped the employees by department name and reported the counts and average salary per department.

| DEPARTMENT | EMP_COUNT | AVG_SALARY |
|------------|-----------|------------|
| ACCOUNTING | 3         | 2916.67    |
| RESEARCH   | 5         | 2175.00    |
| SALES      | 6         | 1566.67    |

If you'd like additional metrics (total payroll, highest salary, etc.) or want the SQL saved/automated, let me know—otherwise this result matches the current `DEMO` schema as shown.

GPT-5.1-Codex-Mini (Preview) • 0.33x



# Implement the following safeguards

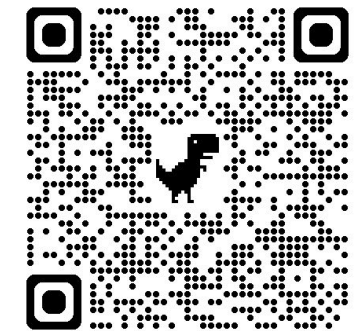
- Assign minimum permissions
- Avoid production database access
- Audit LLM activity



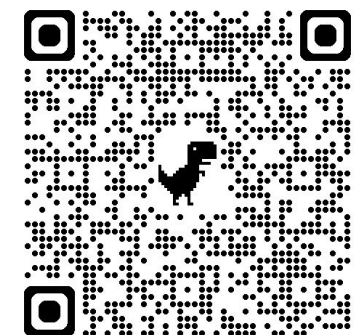


Start by reading those first

[Getting Started with our MCP Server for Oracle Database \(by Jeff Smith\)](#)



[SQLcl MCP Server Official Documentation](#)



# Customizing SQLcl



## Display username & connection name

```
DEMO@dev_demo > select 1 from dual;
```

```
1
```

```
1
```

```
> set sqlprompt "@|blue _USER|@@@|green _CONNECT_IDENTIFIER|@ > "
```



## Display status bar

TERMINAL

DEMO@dev\_demo >

DEMO | dev | NONE | /Users/rg/DEMO/ | None | No time

**USERNAME | GIT | PROJECT | CWD | TRANSACTION STATUS | TIMING**

> set **statusbar** on

> set **statusbar default** username git project cwd txn timing





## set linenumbers on | off

```
DEMO@dev_demo > set linenumbers on
```

```
DEMO@dev_demo > select 'I really like SQLcl!'
2* from dual;
```

```
'IREALLYLIKESQLCL!'
```

```

I really like SQLcl!
```

```
DEMO@dev_demo > set linenumbers off
```

```
DEMO@dev_demo > select 'But I dont like line numbers'
from dual;
```

```
'BUTIDONTLIKELINENUMBERS'
```

```

But I dont like line numbers
```

```
DEMO@dev_demo > |
```



# set termout on | off

Controls wheter to display the results of SQL when running a script

TERMINAL

```
DEMO@dev_demo > set termout on|
```





## set serveroutput on | off

Print or not output generated by DBMS\_OUTPUT

```
DEMO@dev_demo > set serveroutput on
DEMO@dev_demo > exec dbms_output.put_line('Hello!');
Hello!

PL/SQL procedure successfully completed.

DEMO@dev_demo > set serveroutput off
DEMO@dev_demo > exec dbms_output.put_line('No message will be shown');

PL/SQL procedure successfully completed.

DEMO@dev_demo >
```



## set feedback on | off

Controls whether to display the number of records returned/affected by a statement

TERMINAL

```
DEMO@dev_demo > set feedback on|
```







set define on | off | '&' | '#'

Enable / disable prompting

TERMINAL

```
DEMO@dev_demo > sh|
```



# show all params

Display all currently set parameters

```
DEMO@dev_demo > s|
```



Do I need to set those parameters every time?





## Define your login script



Add another path to run custom scripts at SQLcl start

Users/<your\_user>/ .zshrc (or .bash\_profile)

```
export SQLPATH="/Users/rg/apps/sqlcl_scripts/"
```







## Define custom settings only once

 login.sql 1 X

Users > rg > Downloads >  login.sql > ...

```
1 set sqlprompt "@|blue _USER|@@|green _CONNECT_IDENTIFIER|@ > "
2 set linenumbers off
3 set define '&
4 set feedback on
5 set serveroutput on
6 set termout on
7 set sqlformat ansiconsole
8 set statusbar on
9 set statusbar default username git project cwd txn timing
```



**GitHub - rgrzegorzcyk/conferences: A place  
for public content related to my  
presentations at various conferences**

github.com

<https://github.com/rgrzegorzcyk/conferences>



Thank you!  
Feel free to reach me at  
the event or online!



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You can find more at  
[www.pretius.com](http://www.pretius.com)

DEMO@dev\_demo >



ORACLE

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